A Weakly Supervised Bitemporal Scene Change Detection Approach for Pixel-Level Building Damage Assessment Using Pre- and Post-Disaster High-Resolution Remote Sensing Images

Wenfan Qiao, Li Shen, *Member, IEEE*, Wei Wang, and Zhilin Li

***Abstract*— Sudden-onset natural disasters, such as destructive earthquakes, pose significant threats to human life and property. The use of high-resolution remote sensing (HRRS) images for automated assessment of building damage can rapidly and accurately provide spatial distribution information and statistical data on building damage, assisting in disaster response and relief efforts. However, the task is exceedingly challenging due to the diverse and intricate appearance of damaged buildings in HRRS images, coupled with interference from surrounding areas that exhibit certain damage characteristics as a result of the disaster. To overcome these issues, this article proposes a weakly supervised building damage assessment method based on scene change detection in pre- and post-disaster bitemporal images. This method fully leverages visual information of building boundaries and deeper semantic information of building scenes from pre-disaster images to guide the identification of building damage in post-disaster images. Specifically, the method first generates fine-grained subbuilding objects with detailed boundaries from pre-disaster images by combining semantic segmentation of buildings with superpixel segmentation. Then, bitemporal image scene blocks obtained using sub-building objects as clues are input into our proposed Siamese local–global visual transformer (SLgViT) network, enabling scene change detection guided by deep semantic information from pre-disaster images. Finally, the change detection results serve as the basis to depict pixel-level building damage in post-disaster images. The proposed SLgViT network is primarily composed of a specially designed local–global visual transformer (LgViT) module and a cross-Siamese interaction fusion (CSIF) module, both of which play a crucial role in the deep mining and integrated interaction of local and global semantic features from pre- and post-disaster images. It is noteworthy that our method operates in a weakly supervised manner. The training of the SLgViT network requires only scene patches centered around building objects from pre- and post-disaster bitemporal images, along with image-level annotations. Experiments conducted with satellite images from the 2010 Port-au-Prince, Haiti earthquake and unmanned aerial vehicle (UAV) images from the 2019 Changning, China earthquake have demonstrated the effectiveness and superior performance of the proposed method.**

***Index Terms*— Building damage assessment, convolutional neural network (CNN), high-resolution remote sensing (HRRS) image, scene change detection, visual transformer (ViT), weakly supervised semantic segmentation (WSSS).**

I. Introduction

RAPID and accurate assessment of building damage following sudden-onset natural disasters, such as destructive earthquakes, is critical for supporting effective disaster response and relief efforts [1]. Traditional field survey methods for mapping building damage areas can be hazardous, challenging, and inefficient due to harsh environments, limited communication, and complex traffic conditions. In contrast, high-resolution remote sensing (HRRS) observations offer a safe and efficient alternative, enabling accurate surface representations of disaster-stricken areas and providing detailed characterizations of building damage with minimal exposure to dangerous conditions. Consequently, in recent years, a growing number of researchers have been dedicated to developing automated methods for detecting building damage using HRRS images, thus improving the overall performance of building damage assessment [2], [3].

In terms of building damage detection methods, there are primarily two categories based on data differences: those that utilize post-disaster monoporal images and those that integrate both pre- and post-disaster bitemporal images [2], [4]. Compared with the latter, methods using post-disaster monoporal images benefit from easier data acquisition. Within this context, the abundant detail provided by HRRS images plays a crucial role in obtaining information about building damage solely from post-disaster data. Advanced feature extraction methods are essential for capturing the critical details within post-disaster scenes, ensuring the successful application of post-disaster monoporal images in building damage assessment. Currently, deep learning techniques, particularly convolutional neural networks (CNNs) [5], have made significant strides in automatically extracting rich hierarchical features from HRRS images. CNNs

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effectively eliminate the time-consuming and prior knowledge-dependent process of designing hand-crafted features, allowing for more powerful feature representations. Furthermore, in the task of building damage assessment, considering global information are crucial for accurately capturing the interrelations between buildings and complex damage patterns [6]. However, CNN models primarily rely on local receptive fields and convolutional operations, which limits their ability to effectively capture global information. The Transformer [7], an attention-based architecture initially proposed for natural language processing (NLP), is designed to capture longer distance dependencies between objects. Dosovitskiy et al. [8] explored the application of vision transformers (ViTs) in image processing, demonstrating its effectiveness in capturing global relationships within images. This led to the development of specialized ViTs designed specifically for visual image processing, and based on which numerous improved ViT models have emerged [9], [10], [11], [12], [13], [14], [15], [16]. In addition, some researchers have investigated hybrid methods that integrate CNN and Transformer frameworks [17], [18], [19], [20]. These approaches aim to combine the strengths of CNNs in extracting local image features with the global information processing capabilities of Transformers. However, to fully leverage the combined strengths of CNNs and Transformers and realize their complementary advantages, it is essential to explore more effective and deeper integration methods between them [20].

Compared with general objects, damaged buildings in post-disaster images exhibit the following unique characteristics: 1) variable shapes and spectral properties; 2) disorganized and noisy appearance with no clear boundaries; and 3) strong inter-class similarity and intraclass heterogeneity. The presence of these characteristics significantly impacts the effect to identify building damage based on post-disaster image information. However, enhancing the feature extraction capabilities of existing methods is insufficient for fundamentally improving the effectiveness of building damage detection. Consequently, methods that rely solely on post-disaster images for detecting building damage face significant limitations.

With the continuous enrichment of data acquisition channels, we have the capability to obtain a wealth of pre-disaster remote sensing imagery to support post-disaster building damage assessment [2]. When both pre- and post-disaster data are available, the key research breakthrough lies in how to mine and utilize the prior information of pre-disaster imagery to guide building damage detection in complex post-disaster scenes [21]. Currently, the method of utilizing bitemporal imagery for change detection has become widely adopted for extracting areas of building damage [22], [23], [24], [25], [26]. This involves analyzing the spectral and morphological differences presented in images taken before and after a disaster to detect and extract information about building damage. However, change detection results can also introduce many nonbuilding damage-related changes, affecting the accuracy of the building damage detection. Despite this, compared with methods that only use post-disaster data, these approaches can leverage prior knowledge from pre-disaster images to obtain more reliable and accurate building damage assessment results. In addition, pre-disaster imagery and data may provide essential clues for effectively locating candidate areas of building damage in post-disaster images, potentially leading to better results by excluding irrelevant interference [27], [28].

In summary, due to the unique characteristics of damaged buildings, achieving more precise and reliable post-disaster building damage assessments is inseparable from the support of pre-disaster prior information. However, based on the current availability of data, acquiring existing data containing pre-disaster prior information (such as building vector data) remains challenging, while the accessibility of pre-disaster images continues to improve. Therefore, integrating pre- and post-disaster images for building damage detection is a necessary methodological framework to enhance detection accuracy. A critical issue that remains unsolved is how to effectively extract and utilize the prior information from pre-disaster images within this framework. We believe that the key to effectively leveraging prior information from pre-disaster images lies in designing strategies that fully exploit the visual information of building boundaries and the deeper semantic information of building scenes contained within pre-disaster images, and in combining these insights with the building damage characteristics observed in post-disaster images to comprehensively and accurately assess building damage. Furthermore, compared with general objects like buildings, pixel-level annotation of damaged buildings is more time-consuming, expensive, and prone to human error, making it difficult to obtain standardized, high-quality pixel-level annotations [29]. Therefore, pixel-level annotation is not practical for post-disaster building damage detection tasks. In addition, the proportion of damaged buildings is typically small, making it challenging to acquire a large volume of pixel-level annotation samples. In this article, there is an urgent need for weakly supervised methods based on image-level annotations to perform building damage detection.

With the aforementioned considerations in mind, this article introduces a weakly supervised building damage assessment method based on scene change detection in pre- and post-disaster bitemporal images. This method first generates fine-grained sub-building objects with detailed boundaries from pre-disaster images by combining semantic segmentation of buildings with superpixel segmentation. Then, bitemporal image scene blocks obtained using sub-building objects as clues are input into our proposed Siamese local–global visual transformer (SLgViT) network, enabling scene change detection guided by deep semantic information from pre-disaster images. Finally, the change detection results serve as the basis for depicting pixel-level building damage in post-disaster images. Overall, the main contributions of this article are reflected in the following aspects.

1) We propose a novel weakly supervised framework for pixel-level assessment of post-disaster building damage through scene change detection using bitemporal HRRS images captured before and after disasters. This framework integrates building object localization, scene change detection, and pixel-level building damage inference into a unified system. It effectively exploits visual information of building boundaries and deeper semantic

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information of building scenes from pre-disaster images, combining these with post-disaster building damage characteristics for a comprehensive pixel-level damage assessment.

2) We design the SLgViT network for scene change detection, which primarily consists of a specially designed local–global visual transformer (LgViT) module and a cross-Siamese interaction fusion (CSIF) module. These components play a crucial role in mining and integrating deep local and global semantic information from pre- and post-disaster images, facilitating effective utilization of deeper semantic information of building scenes.
3) Our method operates under weak supervision, requiring only pre- and post-disaster bitemporal image scene blocks centered around building objects and image-level annotations. Through pixel-level building damage inference, it transforms scene-level results into detailed pixel-level building damage detection results.

The rest of this article is structured as follows. Section II provides an overview of related literature. Section III introduces the proposed methodology, detailing the processing framework and its main components. Section IV presents our experiments and the comparative analysis of results. Section V contains a comprehensive discussion. Finally, the conclusion is drawn in Section VI.

II. Related Works

Due to the advantages of noncontact and macroscopic monitoring, the remote sensing technology has facilitated a considerable amount of research on post-earthquake building damage detection. Studies have involved various data sources, such as optical remote sensing [6], [21], [30], [31], [32], synthetic aperture radar (SAR) [4], [33], [34], and light detection and ranging (LiDAR) point clouds [35], [36], [37]. Here, we focus on building damage assessment methods based on HRRS images.

A. Using Only Post-Disaster HRRS Images

The key to building damage assessment based on post-disaster HRRS images is constructing an appropriate feature representation to effectively distinguish between damaged building areas and other background regions in the feature space. Li et al. [38] proposed an unsupervised method based on a modified Chinese Restaurant Franchise (CRF) model, which detects earthquake-induced roof holes in unmanned aerial vehicle (UAV) imagery by fusing color and shape features. Moya et al. [39] created a set of features by constructing 3-D gray-level co-occurrence matrices, which were then fed into a support vector machine classifier to detect collapsed buildings. Some researchers have explored the relationship between buildings and their cast shadows in remote sensing images, which can serve as clues for infer damaged areas [40], [41]. Furthermore, high-level semantic features based on the bag-of-words model have been designed for building damage detection [42], [43]. Overall, these traditional methods (also referred to as manual feature design approaches) still exhibit significant shortcomings, which primarily include reliance on the extensive prior knowledge for feature design, the imperfection of manually designed features, and the difficulty in transferring feature rules.

CNNs [5], as powerful deep learning structures, are capable of automatically extracting rich hierarchical features from images, demonstrating immense potential in the task of automated assessment of building damage in remote sensing. Bai et al. [44] employed the SqueezeNet network [45] to distinguish between built-up and nonbuilt-up areas. Subsequently, they developed a modified wide residual network to determine the extent of building damage. Gebrehiwot et al. [46] further utilized the visual geometry group (VGG) architecture to classify drone imagery acquired after flood disasters. Yang et al. [47] employed a variety of CNN models combined with transfer learning methods to identify buildings damaged by of global information. To obtain more accurate information on the location of building damage, the weakly supervised semantic segmentation (WSSS) method based on class activation maps (CAMs) [48], [49] has been employed to assist in pixel-level building damage detection tasks [6], [50]. However, the blurred boundaries and messy textures of damaged buildings in post-disaster HRRS images could have a negative impact on obtaining high-quality CAMs. In addition, Song et al. [51] utilized the DeepLab v2 semantic segmentation model for initial pixel-level identification of damaged building areas, followed by the use of superpixel segmentation to refine the boundaries of the damaged structures.

In the task of building damage assessment, the consideration of global information is crucial for accurately capturing the relationships between buildings and complex patterns of damage. However, CNN models primarily rely on local perception and convolutional operations, which limit the effective capture of global information. This may lead to difficulties for the model in understanding long-range dependencies, thereby impacting the model's performance. On the other hand, given the success of Transformers [7] in speech recognition and machine translation, Transformer-based networks have also made their way into the field of computer vision. The advent of the ViT model [8] has led to the development of image classification architectures using Transformers. The self-attention mechanism of Transformers allows the model to concurrently focus on all positions within an image, thus better understanding the global distribution and interaction relationships of objects [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19]. The T2TViT model, as proposed by Yuan et al. [12], employs an innovative hierarchical token transformation approach to enhance local feature representation while simultaneously maintaining a focus on global features. This approach is supported by a backbone structure reminiscent of ResNet [52], significantly improving the encoding efficiency of the primary network. Furthermore, researchers have explored methods that combine CNN and Transformer frameworks, aiming to integrate CNN's capability for extracting local image features with Transformer's ability to focus on the global information [17], [18], [19]. Tu et al. [19] introduced the MaxViT model, which combines CNNs and Transformers to create fundamental encoding blocks. By stacking these blocks, the model forms a layered visual backbone capable

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of enabling global–local spatial interactions across images of varying resolutions.

Therefore, building damage identification methods based on Transformers are increasingly gaining attention [20], [53], [54]. Cui et al. [53] proposed an improved Swin Transformer for segmenting dense urban buildings from post-disaster remote sensing images with complex backgrounds. This method effectively categorized and identified collapsed and noncollapsed buildings from remote sensing images of the Yushu and Beichuan earthquakes. Through ablation experiments and comparative studies, they also validated the effectiveness and superiority of the proposed method relative to the original Swin Transformer and several mature CNN-based segmentation models. Kaur et al. [54] introduced a novel Transformer-based network for assessing building damage. The network leverages multiresolution hierarchical spatial features and captures temporal variations in the feature domain after processing spatial features with transformer encoders. Testing on the large-scale xBD dataset demonstrated that this method achieved state-of-the-art performance in building damage assessment. Shi et al. [20] proposed the cross-conv-transformer model, which utilizes CNN's inherent advantages, such as sensitivity to local image information, to enhance feature extraction for building damage scenes. At the same time, it fully leverages Transformer's strengths, such as parallel computation, global perspective, and flexible feature stacking.

B. Using Both Pre- and Post-Disaster Data

With the continuous enrichment of data acquisition channels, we now have the capability to obtain more pre-disaster remote sensing imagery and geographic information data to support post-disaster building damage assessment. Compared with methods that solely use post-disaster data, introducing prior knowledge from pre-disaster data can yield more reliable and accurate building damage assessment results [55]. To precisely locate building boundaries and eliminate interference from other objects, Janalipour and Mohammadzadeh [56] and Ye et al. [57] adopted statistical learning methods, utilizing multiple features (such as geometry, texture, and gradient features) to classify building damage in post-disaster images, aided by pre-disaster building vector maps. Ji et al. [27] used an improved SqueezeNet, combined with pre-disaster building footprint data, to identify damaged buildings in post-disaster images. Ge et al. [58] proposed a new framework that utilizes pre-disaster building distribution data and post-disaster panchromatic imagery to rapidly identify collapsed building objects. The aforementioned studies prove that pre-disaster auxiliary information does enhance the accuracy of building damage identification to some extent. However, the additional geographic information aid data are not always available, which limits its applicability in rapid disaster emergency response.

Currently, utilizing pre-disaster and post-disaster remote sensing images for building damage assessment has become an alternative strategy, with methods based on remote sensing image change detection being one of the most commonly used approaches at present [22], [23], [24], [25], [26]. The core idea is to locate and detect damage information based on the spectral and morphological differences presented in remote sensing imagery at different times. Considering the practical needs of emergency rescue and post-disaster reconstruction, Qing et al. [59] proposed a hierarchical workflow for building damage assessment. This workflow employs CNNs for direct remote sensing change detection at the superpixel level. Experimental results from case studies demonstrated that the proposed workflow can effectively and accurately locate and classify damaged buildings. Lu et al. [60] introduced a dual-temporal cross-attention module within the change detection framework, aiming to effectively capture the spatiotemporal semantic relationships between pre- and post-disaster remote sensing images, achieving high-precision damage detection on the disaster dataset xBD. Zheng et al. [22] proposed an object-based semantic change detection framework, where deep object localization networks and damage classification networks are integrated into a unified semantic change detection network for end-to-end assessment of building damage. However, change detection methods may introduce noise from changes unrelated to building damage, suggesting that change detection results might necessitate further postprocessing for optimization. In addition, obtaining the reliable and accurate building damage results remains challenging due to the significant differences in color, spectral characteristics, and viewpoints between pre-disaster and post-disaster images.

In addition to these approaches, Bai et al. [61] employed a fully supervised semantic segmentation network, U-Net [62], in conjunction with pre- and post-disaster HRRS imagery overlay data, to perform pixel-level semantic segmentation of building damage. Experiments indicate that this method can achieve commendable results in detecting building damage. Nonetheless, this approach requires high-quality, pixel-level semantic annotations of building damage, the production of which is time-consuming and labor-intensive. It also demands highly specialized annotators, making the process for creating samples a limitation in the practical application of this method.

III. Methodology

In this section, we will present the proposed weakly supervised bitemporal scene change detection approach for pixel-level building damage assessment using pre- and post-disaster HRRS images in detail. The overall framework of the proposed approach is shown in Fig. 1, which mainly consists of three components: 1) sub-building object generation; 2) bitemporal sub-building scene change detection; and 3) pixel-level building damage inference. A comprehensive explanation of these three components is provided in this section.

A. Sub-Building Object Generation

1) Pre-Disaster Building Extraction: After a disaster, it is not only buildings that may suffer damage, but also other objects could be affected and may exhibit characteristics similar to those of damaged buildings. In addition, rubble from damaged buildings might be scattered across roads, making it difficult to distinguish between the actual building damage and other types of debris in post-disaster HRRS images. As a result, extracting building damage from the complex

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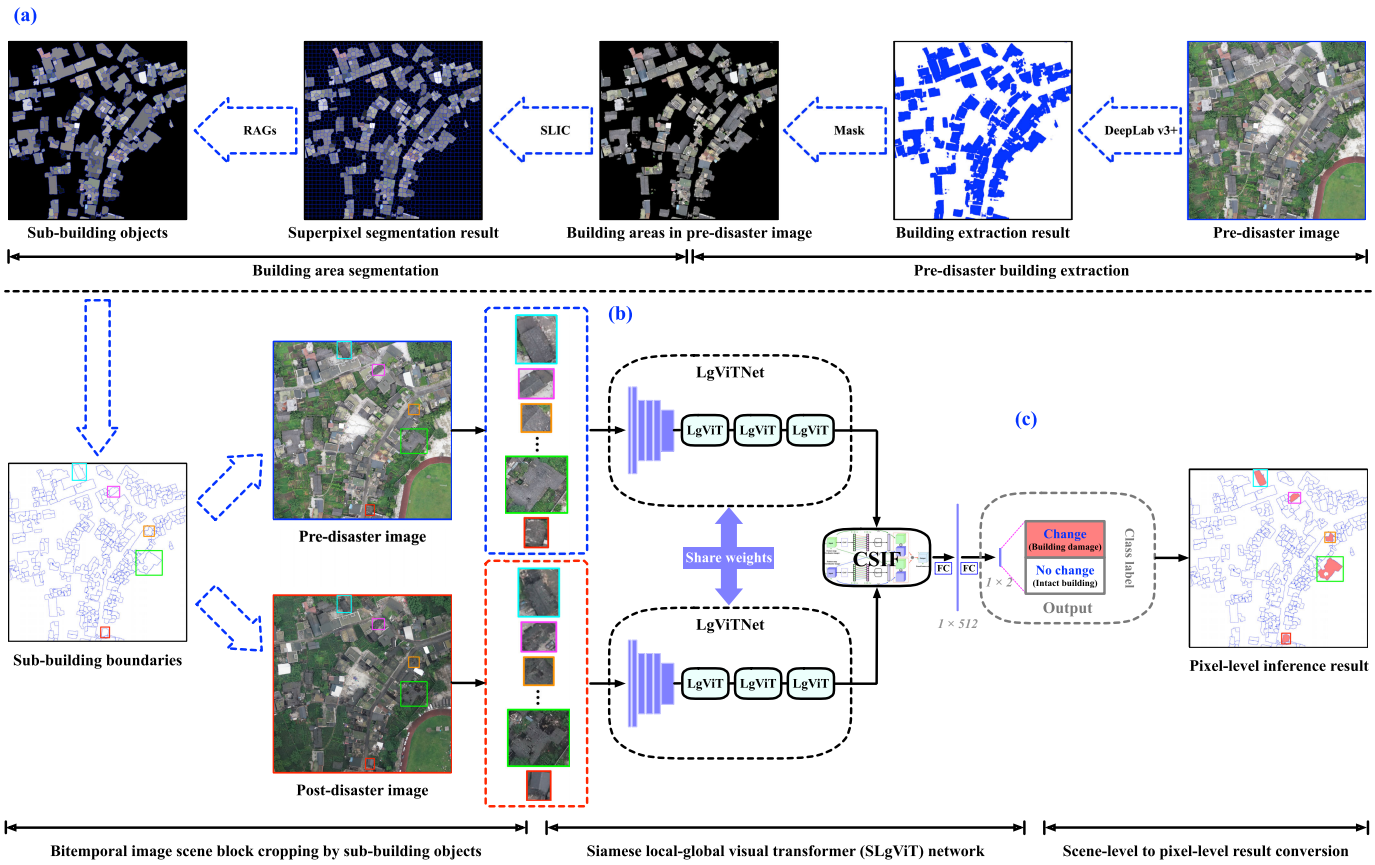


Fig. 1. Overall framework of the proposed building damage detection method. (a) Sub-building object generation. (b) Bitemporal sub-building scene change detection. (c) Pixel-level building damage inference.

post-disaster HRRS images poses a considerable challenge. However, it is important to note that building damage can only occur within the areas occupied by buildings before the disaster. Therefore, information of pre-disaster building area locations can be used to eliminate interference from other ground objects. In comparison with damaged buildings, most buildings exhibit regular interior structures, leading to higher interclass heterogeneity and lower intraclass heterogeneity in their appearance on HRRS images.

Through the above analysis, in our proposed framework, building areas are first extracted from pre-disaster images. This step is crucial to eliminate interference from nonbuilding areas. It is important to note that the required pre-disaster building area location information only needs to be broadly accurate. In other words, even if the building extraction results are not very detailed, the overall impact on our approach will not be noticeable, since the subsequent building segmentation is able to generate sub-building objects with detailed boundaries from building areas. In this step, most current fully convolutional network (FCN)-based semantic segmentation methods can satisfy the required accuracy for pre-disaster building extraction. Therefore, the design and improvement of building extraction methods are not our primary focus.

DeepLab v3+ [63], a prevalent semantic segmentation network, is widely used for building extraction tasks in HRRS images. This network, featuring an encoder–decoder structure and an atrous spatial pyramid pooling mechanism, is adept at capturing valuable multiscale information from various feature map scales. Therefore, we select the DeepLab v3+ semantic segmentation network with a ResNet-101 [52] backbone to extract building areas from pre-disaster HRRS images, as shown in Fig. 1(a). To further enhance the performance of the DeepLab v3+ model in building extraction, we initially pretrained the model on the Inria aerial image labeling dataset [64], which is publicly available. The model is then fine-tuned using our pixel-level building annotation data from pre-disaster earthquake zones. This method ensures comprehensive training of the model, enabling it to achieve exceptional performance in extracting buildings in pre-disaster scenarios.

2) Building Area Segmentation: As illustrated in Fig. 1(a), the pre-disaster building extraction outcomes derived from semantic segmentation methods are often incapable of accurately delineating the boundaries between adjacent structures. Consequently, relying solely on these building extraction results with blurred boundary information is not sufficient to provide detailed guidance and prior information support for achieving fine-grained detection of each individual building's damage or even the partially damaged areas of buildings. To address this issue, we devise a targeted strategy that effectively delineates more detailed sub-building objects, providing enhanced boundary information for building damage detection.

To increase the richness of building area boundary information, it is essential to first eliminate interference from

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areas outside the buildings. Therefore, utilizing the extracted building area results to create a mask for pre-disaster images helps filter out nonbuilding areas in the images. Subsequently, refining the segmentation of building areas on the mask image is achieved through a superpixel segmentation algorithm. However, as depicted in Fig. 1(a), superpixel segmentation often leads to the generation of a large number of excessively fragmented building segments. These segments fail to distinctly manifest prominent building features and are unsuitable for serving as detection units for building damage. In addition, an excessive number of segments also significantly increases computational costs. To address this issue, we merge adjacent fragmented segments into more complete and cohesive units based on specific similarity rules after the superpixel segmentation process. While the merged homogeneous units may not represent individual buildings, they exhibit features of buildings, and we refer to these as sub-building objects. Notably, our methodological framework employs simple linear iterative clustering (SLIC) [65] superpixel segmentation algorithm and region adjacency graphs (RAGs) [66] to execute the segmentation and merging of fragmented segments into homogeneous units. The choice of these algorithms is flexible and can be replaced by any suitable superpixel segmentation and merging methods.

It is worth noting that our segmentation strategy is fundamentally different in purpose from conventional object-oriented segmentation methods. Object-oriented segmentation, whose primary objective is to delineate each individual building more effectively, often requires selecting the optimal segmentation scale to ensure the method's effectiveness. However, selecting the optimal segmentation scale necessitates careful consideration of the actual image scene conditions, a task that is both time-consuming and complex. Our segmentation strategy mitigates the issue of selecting the optimal scale because it does not require obtaining complete, standalone building objects through optimal segmentation. Instead, it focuses on generating basic processing units that reflect the characteristic features of buildings. Even if a building is segmented into sub-building objects through different scales, the detailed boundary information of these sub-building objects ensures that the final damage detection results for the building will not be significantly affected. Thus, in our segmentation strategy, as long as the merging similarity threshold in the RAGs is set within an appropriate range (ensuring that the segmented units are at the sub-building level), it can fulfill the conditions for generating the required sub-building objects.

B. Bitemporal Sub-Building Scene Change Detection With SLgViT Network

In the overall methodological framework of the proposed building damage detection method, the use of the designed SLgViT network for bitemporal building scene change detection is a crucial part in determining whether buildings have been damaged after a disaster. The SLgViT network, with its Siamese dual-branch structure, accommodates input from both pre-disaster and post-disaster bitemporal image scene blocks. By fusing features from these bitemporal image scene blocks, it performs change detection and thus determines building damage, as shown in Fig. 1(b).

The training data for the SLgViT network are based on image-level annotations of paired building image scene blocks from before and after disasters. If a building in the post-disaster scene is damaged, this pair of image scene blocks are marked as "change"; otherwise, it is marked as "no change." This annotation considers the following points.

1) The selection of our training samples is based on building-centric image scene blocks, where the main ground objects are buildings. This choice of image scenes effectively eliminates false change detections caused by variations in other ground objects.
2) Change detection performed by the SLgViT network, which integrates high-level semantic features from bitemporal image scene blocks, can eliminate interference caused by minor shifts in building targets between pre-disaster and post-disaster images.
3) Changes due to the normal construction or demolition of buildings generally require a longer time period, and change detection in building scenes based on a shorter interval between pre-disaster and post-disaster images can ignore interference from such situations.

Considering these points, we believe the changes in the paired building image scene blocks from before and after disasters are due to building damage, thus establishing a correlation between bitemporal scene change detection results and the condition of the building damage, where "change" indicates building damage and "no change" signifies an intact building.

In carrying out our method in test images, the position information of generated sub-building objects is used as clues to acquire pre-disaster and post-disaster bitemporal sub-building image scene blocks as input to the trained SLgViT network. The bitemporal sub-building image scene blocks are obtained by cropping from corresponding pre-disaster and post-disaster test images based on the minimum bounding rectangles of each generated sub-building object, as shown in Fig. 1(b). These paired pre-disaster and post-disaster sub-building image scene blocks match the training data of the SLgViT network. That is, conducting change detection on these bitemporal sub-building image scene blocks with the trained SLgViT network can yield the damage status of sub-building objects. Compared with building damage detection relying solely on post-disaster information from the image scene blocks, our proposed SLgViT network can leverage deep semantic prior knowledge of pre-disaster building scenes to achieve superior performance. The proposed SLgViT network primarily consists of the LgViT network (LgViTNet) and CSIF modules, with LgViTNet responsible for extracting and representing high-level semantic features from pre-disaster and post-disaster bitemporal image scene blocks, and the CSIF module for comprehensive interaction and fusion of the obtained pre- and post-disaster features.

1) LgViTNet: In the bitemporal scene change detection task, integrating global feature relationships across different locations in the scene facilitates accurate judgment of scene changes. Simultaneously, the importance of perceiving local features cannot be overlooked, as certain local regions may be

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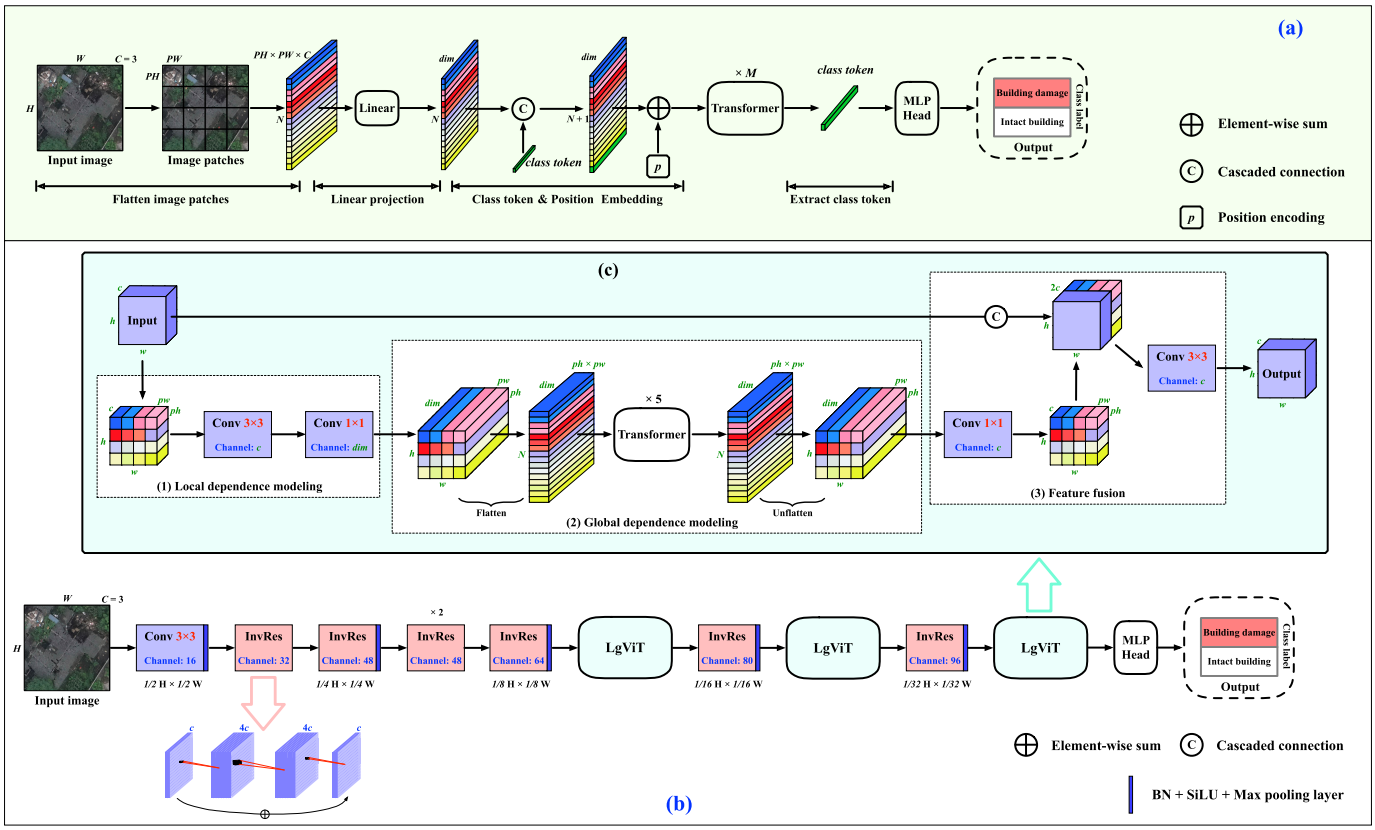


Fig. 2. Architectures of the proposed LgViTNet. (a) Visual transformer (ViT). (b) LgViTNet. (c) LgViT module.

focal points of scene variation, with their information playing a decisive role in change detection outcomes. Considering the complementary relation of local and global features, we design a shared feature encoder network, LgViTNet, to capture both local details and long-range dependencies between features, thus achieving comprehensive perception for pre- and post-disaster bitemporal image scene features.

To learn global representations, employing ViTs with the global vision processing capabilities can be crucial. For a standard ViT network, as depicted in Fig. 2(a), the input image $I \in \mathbb{R}^{C\times H\times W}$, with height ($H$), width ($W$), and ($C$) channels, is first divided into image patches. Each image patch is then flattened into a 1-D vector $I_{1-D} \in \mathbb{R}^{1\times(PH\times PW\times C)}$, and these 1-D vectors are treated as tokens, which are concatenated to form a 2-D matrix $I_{2-D} \in \mathbb{R}^{N\times(PH\times PW\times C)}$. This matrix is then projected into a fixed-dimensional space $I_p \in \mathbb{R}^{N\times \dim}$. After incorporating the class token and position embeddings, the network learns interpatch representations through M transformer blocks. Here, PH, PW, and N, respectively, denote the height, width, and number of divided image patches. Despite the advantage of ViTs in representing global features, they lack the image-specific inductive biases inherent to CNNs, often overlooking CNN's inherent sensitivity to local features. Therefore, combining CNNs and Transformers is considered a strategy that leverages the complementary strengths of both. The current mainstream methods of integration involve convolutional tokenization through CNNs combined with Transformers [17] and direct stacking connections [19]. However, these methods do not achieve a deep integration but only a superficial merger. Thus, designing a new way to realize the deep complementary advantages of CNNs and Transformers is a critical issue.

Our proposed LgViTNet primarily consists of inverse residual (InvRes) blocks [67] and LgViT modules, as illustrated in Fig. 2(b). Here, the LgViT module signifies a novel method for the integration of CNNs and Transformers, differentiating itself from the general methods of combining CNNs and Transformers. LgViT ingeniously inserts transformer operations during the convolutional process. This strategy uses Transformers as a means of learning global representations, implicitly integrating the attributes of CNNs and Transformers into the network. Such integration fosters a profound complementarity between CNNs and Transformers. The LgViT module contains local dependence modeling, global dependence modeling, and feature fusion for the input feature, with the specific structural details depicted in Fig. 2(c). For a given input feature map $F \in \mathbb{R}^{c\times h\times w}$, with height, width, and number of channels denoted by h, w, and c, respectively, the LgViT module initially divides it into a sequence of feature patches with height ph, width wh, and c channels. During the local dependence modeling stage, LgViT applies a 3×3 convolutional layer followed by a 1×1 convolutional layer, which encodes the local spatial information of the input features and projects the feature map into a fixed-dimensional space $F_l \in \mathbb{R}^{\dim\times h\times w}$. Subsequently, to perform global dependence modeling, F_l is flattened into N 2-D matrices $F_{2-D} \in \mathbb{R}^{\dim\times(ph\times pw)}$ based on the previous division of feature patches. These matrices are then concatenated to form a 3-D matrix

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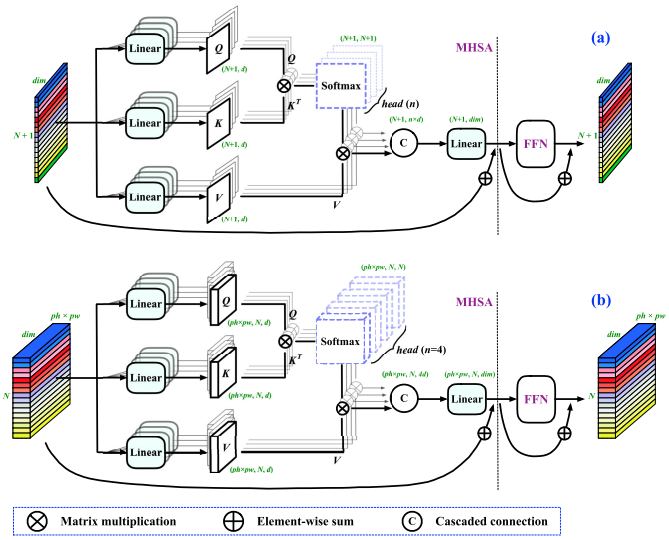


Fig. 3. Transformer block. (a) Transformer block in the standard ViTs. (b) Transformer block in the proposed LgViT module.

$F_{3-D} \in \mathbb{R}^{(ph\times pw)\times N\times \dim}$. Here, $N = (h \times w)/(ph \times pw)$ is the number of feature patches. The F_{3-D} are fed into five transformer blocks to encode interpatch relationships and obtain global feature representation $F_g \in \mathbb{R}^{(ph\times pw)\times N\times \dim}$. we then can unflatten the F_g and transform it into a local–global feature space $F_{lg} \in \mathbb{R}^{c\times h\times w}$, achieving both local and global feature perception. Finally, the output of the LgViT module is obtained by fusing F_{lg} and F through cascaded connection operation and a 3×3 convolution.

For each transformer block in the LgViT module, as depicted in Fig. 3(b), it consists of a multihead self-attention (MHSA) module and a feed-forward network (FFN). Similar to the standard MHSA module [Fig. 3(a)], three linear layers are employed for query (Q), key (K), and value (V) computation, but they handle different feature dimensions. Mathematically, for a given input feature map F, the self-attention result for one head $h_i, i \in [1, 2, \ldots, n]$ in the MHSA, denoted by A_{h_i} is calculated as follows:

$$A_{h_i}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}}\right)V \tag{1}$$

where d denotes the dimension of each head in MHSA and $\sqrt{d}$ denotes the scaling factor. The output of MHSA, denoted by M, can be obtained by concatenating the results of each head, calculated as follows:

$$M(F) = F + \text{LN}(\text{Concat}[A_{h_1}, A_{h_2}, \ldots, A_{h_n}]). \tag{2}$$

In the proposed LgViT module, we set the number of heads (n) to 4. The output of the transformer block, denoted by T, can be computation as follows:

$$T(F) = M(F) + \text{FFN}(M(F)). \tag{3}$$

It should be noted that, through the LgViT module, each pixel in the image scene can establish relationships with all other pixels, as illustrated in Fig. 4. In this example, the red pixel forms connections with the green pixel through Transformers, while the green pixel has already encoded information about neighboring blue pixels using convolution.

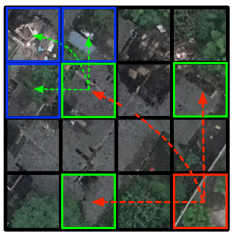


Fig. 4. Example of image dependence relations from local to global.

Consequently, the red pixel can attend to information from all pixels in the image.

2) CSIF Module: After encoding features separately for pre-disaster and post-disaster bitemporal image scene blocks using the shared LgViTNet, it is necessary to establish connections between them for deep-level feature interaction to achieve change detection in the scene [68], [69]. However, the cascaded connection fusion at the feature map level increases feature complexity, while significantly enhancing attention to the similar parts between feature maps, thereby weakening the representation of differential features caused by object changes. In addition, the process of fusing feature maps through elementwise subtraction leads to an excessive amplification of feature differences, increasing the model's sensitivity to noise. To effectively fuse and leverage the advanced semantic features derived from pre-disaster and post-disaster image scene blocks, we introduce the CSIF module, as illustrated in Fig. 5. This module initially applies channelwise attention [70] to both pre-disaster and post-disaster feature maps, effectively filtering out a substantial amount of irrelevant interference while simultaneously enhancing the focus on significant feature information and details. Following this, it facilitates the comprehensive utilization of pre-disaster and post-disaster features through an attention-based CSIF. Consequently, the proposed CSIF module allows for acute attention to be paid to the feature differences in pre- and post-disaster paired building image scene blocks while filtering out noise interference caused by minor feature disparities, resulting in improved bitemporal scene change detection performance.

Initially, given input feature maps $F_{pre} \in \mathbb{R}^{c\times h\times w}$ and $F_{post} \in \mathbb{R}^{c\times h\times w}$, aggregated features $f_{pre} \in \mathbb{R}^{1\times 1\times c}$ and $f_{post} \in \mathbb{R}^{1\times 1\times c}$ are obtained through global average pooling operations. Here, h, w, and c represent the height, width, and number of channels of the input feature maps, respectively. F_{pre} and f_{pre} denote the features of pre-disaster images, while F_{post} and f_{post} represent features of post-disaster images. Then, f_{pre} and f_{post} are subjected to 1-D convolutions to achieve local cross-channel feature interaction. The coverage range of interaction can be adjusted by setting the kernel size of the convolution. In Section IV, we chose a kernel size of 3 for the 1-D convolution to ensure the local density of cross-channel feature interaction. Subsequently, sigmoid functions are applied to obtain channel weight matrices $W_{pre} \in \mathbb{R}^{1\times 1\times c}$ and $W_{post} \in \mathbb{R}^{1\times 1\times c}$ for the pre-disaster and post-disaster image features, respectively. Finally, the pre-disaster and post-disaster image features interact and fuse in the Siamese network based on W_{pre} and W_{post}. The fused feature

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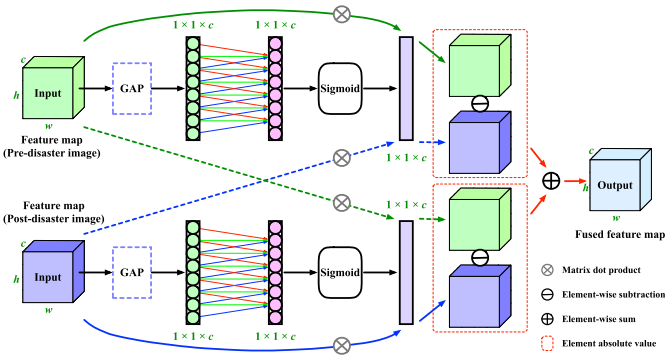


Fig. 5. Proposed CSIF module.

F_{fused} is obtained through the following computation:

$$F_{\text{fused}} = \left|F_{\text{pre}} \cdot W_{\text{pre}} - F_{\text{post}} \cdot W_{\text{pre}}\right| + \left|F_{\text{pre}} \cdot W_{\text{post}} - F_{\text{post}} \cdot W_{\text{post}}\right|. \quad (4)$$

Through this cross-Siamese interaction, the proposed CSIF module can assist change detection models in accurately highlighting the most valuable information related to feature changes in pre- and post-disaster image scene blocks, with the comprehensive utilization of this information effectively enhancing bitemporal scene change detection performance.

C. Pixel-Level Building Damage Inference

To achieve pixel-level building damage inference, we design a strategy to convert image-level damage annotations to pixel-level damage annotations for sub-building objects, as depicted in Fig. 1(b). By employing change detection on pre- and post-disaster bitemporal image scene block using the proposed SLgViT network, we obtain image-level labels that represent the changing status of the sub-building object within the scene, indicating whether sub-building object is damaged. Through this change detection approach, in the test images, all sub-building objects derived from the building areas receive an image-level damage annotation, indicating whether the corresponding sub-building object is damaged. Following the acquisition of damage status detection information for sub-building objects, we determine the pixels contained within each sub-building object based on its boundary information in the images. If a sub-building object is detected as damaged, all pixels within it are marked as damaged; otherwise, they are not marked. This method converts the image-level damage annotations of sub-building objects to pixel-level annotations within them. Upon completing the processing of all sub-building objects generated from the test images, pixel-level building damage detection results can be obtained. The pixel-level results, achieved through this conversion strategy, exhibit visually smooth and uniform interiors with clear and accurate boundaries. Moreover, this method avoids detecting nonbuilding areas within the images, and the segmentation of image scene blocks based on sub-building objects prevents computational redundancy caused by overlapping scene areas, significantly enhancing the method's efficiency.

It should be noted that for general pixel-level inference, due to CPU memory limitations, large-scale HRRS images typically require segmentation into smaller image blocks as processing units for pixel-wise detection. Stitching these inference results of image blocks back together can lead to uneven positions across the entire image result. Compared with existing methods, our pixel-level building damage inference approach overcomes the challenges posed by the segmentation and stitching of HRRS images, simultaneously enhancing the efficiency of pixel-level building damage detection. Overall, the proposed weakly supervised bitemporal scene change detection method provides a more effective and efficient solution for pixel-level building damage assessment in large-scale HRRS images.

IV. Experiments

A. Datasets and Experimental Setup

1) Datasets: To evaluate the effectiveness of the proposed method, we collect HRRS images from two earthquake-affected areas, including images taken before and after the disasters. The data comprise: 1) Satellite images of Port-au-Prince, Haiti, with a resolution of 0.28 m, captured on August 26, 2009 (pre-disaster) and January 13, 2010 (post-disaster) and 2) UAV images of Changning, China, with a resolution of 0.06 m, taken on August 18, 2018 (pre-disaster), and June 18, 2019 (post-disaster), as illustrated in Fig. 6. These two areas sustained significant damage during the earthquakes that occurred on January 12, 2010, and June 17, 2019, respectively. Two datasets based on imagery data corresponding to the two areas are created for building damage detection: the Haidi dataset and the Changning dataset. In each dataset, we select three image patches from pre- and post-disaster images as test areas, each measuring 3500 × 3500 pixels, and conduct pixel-level annotations for both pre-disaster buildings and post-disaster building damage, as marked by the red rectangles in Fig. 6. In addition, nine pre-disaster image patches of the same size are chosen from each dataset for pixel-level building annotations, which are used for training and validating the pre-disaster building extraction model, as indicated by the blue rectangles in Fig. 6.

The details of pixel-level annotations for pre-disaster buildings and post-disaster building damage are shown in Fig. 7. We conduct detailed pixel-level annotations of buildings based on footprint boundaries in the pre-disaster images [29], [71], as indicated in Fig. 7(a). For the pixel-level annotations of post-disaster building damage, we start by comparing pre- and post-disaster images to accurately assess each building and determine whether it has been damaged. Upon identifying a damaged building, we annotate the entire damaged building or specific damaged regions at the pixel level based on the building boundaries from the pre-disaster images. This method helps avoid issues caused by the blurred boundaries of damaged buildings in the post-disaster images. Finally, we perform at least three rounds of cross checking for each annotated image patch to correct inaccurate and erroneous annotations.

In addition, image-level annotation of bitemporal image scene blocks is required for the weak supervision training of the proposed method for building damage detection. We obtain bitemporal image scene blocks by cropping various building object scenes from matched pre- and post-disaster bitemporal HRRS images and annotating them at the image level,

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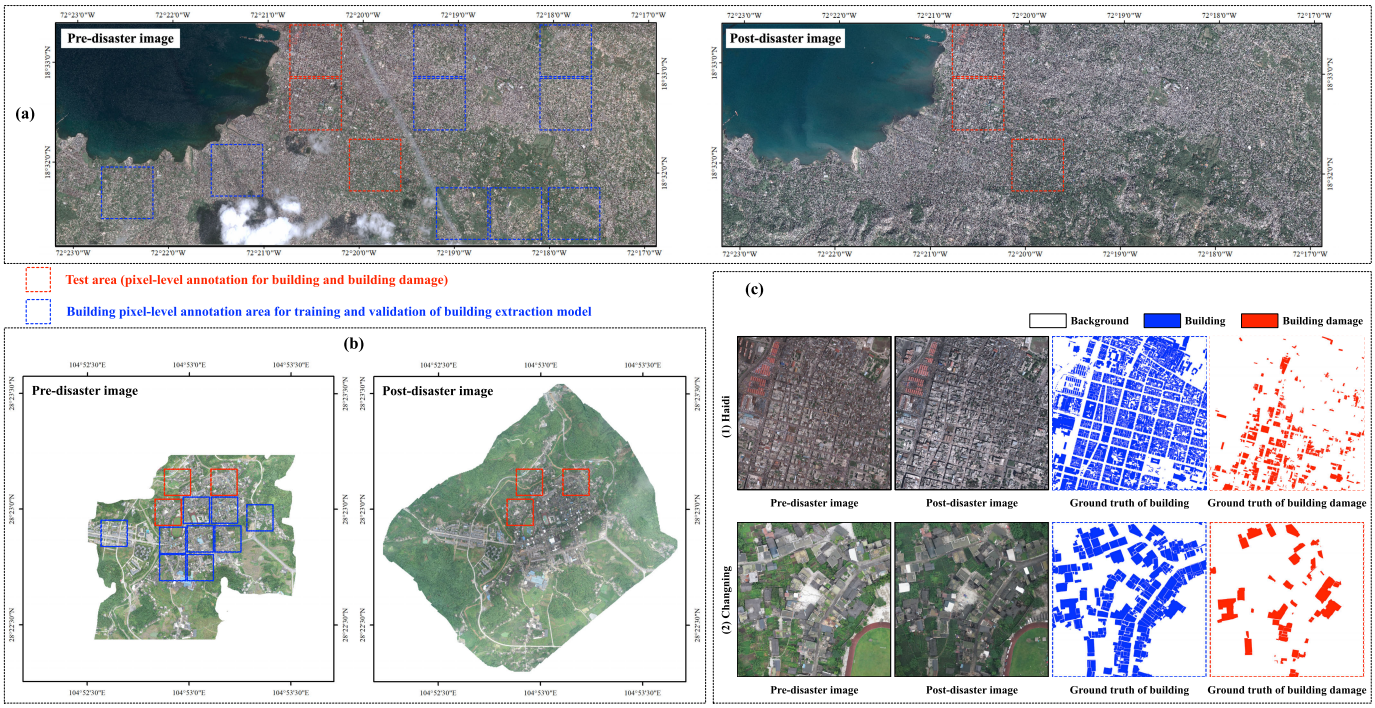


Fig. 6. Demonstration of experimental areas and pixel-level annotation examples of building and building damage. (a) Haidi. (b) Changning. (c) Pixel-level annotation examples of building and building damage in the test area.



Fig. 7. Details of pixel-level annotations for pre-disaster buildings and post-disaster building damage. (a) Annotations of pre-disaster buildings. (b) Annotations of post-disaster building damage. (c) Comparison of pre and post-disaster images for building damage.

as depicted in Fig. 8(a). Bitemporal image scene blocks with building damage in the post-disaster image scene are categorized as "change," while other bitemporal image scene blocks are labeled as "no change." It is important to note that the bitemporal image scene blocks used for model training and validation are acquired from pre- and post-disaster images outside the test areas. All bitemporal image scene blocks are resampled to 256 × 256 pixels before being input into the change detection model, with the detailed data distribution presented in Table I. The image-level annotation data created can also be applied to general scene-level classification methods, where only the post-disaster image scene blocks are utilized.

The categorization of building damage follows five grades according to the European Macroseismic Scale (EMS-98) [72], with an introduction to each building damage grade shown in Fig. 8(b). However, it is essential to acknowledge that remote sensing imagery reflects the spectral response of the surface of objects and can only capture the external condition of buildings, not their internal structure. The presence of debris scattered on or around buildings can alter the spectral readings, significantly impacting the discernment of slight building damage (such as superficial cracks or missing roof tiles of buildings) in HRRS images. Hence, we define building damage in HRRS images as corresponding to grades G3–G5 [73].

2) Implementation Details: We implement all neural network models using PyTorch version 1.10 on a server equipped with an NVIDIA GTX 1080Ti GPU, which features 11.5 GB

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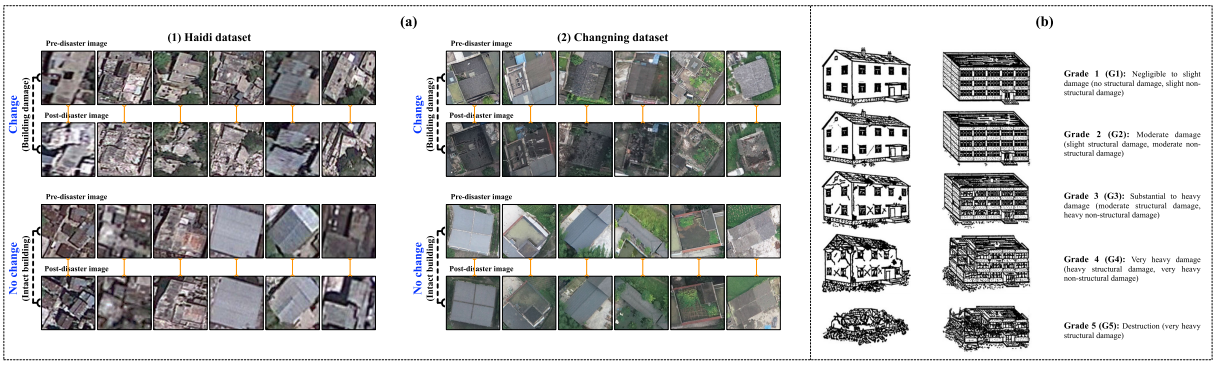


Fig. 8. Image-level annotation examples of bitemporal image scene blocks for change (building damage) and no change (intact building) in Haiti and Changning datasets. (a) Image-level dataset. (b) Building damage grade.

TABLE I
DETAILED DATA DISTRIBUTION FOR IMAGE-LEVEL ANNOTATION OF PRE- AND POST-DISASTER BITEMPORAL IMAGE SCENE BLOCKS

Data	Image size (pixels)	Number of bitemporal image blocks			
		Training and validation		Test	
		Change	No change	Change	No change
Haidi	256 × 256	1700	1700	300	300
Changning	256 × 256	424	930	105	233

of RAM. The training scheme is configured based on empirical determinations, with the number of epochs and batch size set at 100 and 10, respectively. For parameter optimization, we employ stochastic gradient descent (SGD) in conjunction with the cross-entropy loss function. The SGD momentum is fixed at 0.9. The learning rate is initially set at 0.001 and adjusted using the ReduceLROnPlateau strategy [74]. This dynamic learning rate adjustment technique reduces the learning rate when improvement in a designated metric ceases. In our experiments, we monitor the validation loss using the "min" mode. The plateau threshold is set at 0.0001, indicating the threshold for considering a plateau. The reduction factor is set to 0.5, meaning that the learning rate is halved upon detecting a plateau. The patience parameter is set to 3, representing the number of epochs to wait before reducing the learning rate after a plateau is detected. All model inputs are resized to a uniform size of 256 × 256 pixels. To ensure a fair comparison, all building damage detection models included in this study are subjected to identical base settings, with no pretraining processing applied.

3) Evaluation Metrics: We assess the outcomes of the building damage detection using various evaluation metrics, including precision (P), recall (R), F_1-score (F_1), mean intersection over union (MIoU), and overall accuracy (OA). For pixel-level building damage detection, we classify building damage as the positive class and the background as the negative class. For scene-level building damage detection, we classify the building damage scene category as the positive class and the intact building scene category as the negative class. The evaluation metrics are defined as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{5}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{6}$$

$$F_1 = 2 \times \frac{\mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{7}$$

$$\mathrm{MIoU} = \frac{1}{2}\left(\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FN} + \mathrm{FP}}\right) \tag{8}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}}. \tag{9}$$

Here, TP and FP represent the true positive and false positive, respectively, while TN and FN denote the true negative and false negative.

B. Accuracy Assessment of the Proposed Method

The comprehensive detection results of building damage across a wide disaster-affected area can better reflect the reliability and accuracy of the proposed method. Therefore, we conduct both qualitative and quantitative analyses on the overall building damage detection results in the test areas to verify the performance of the proposed method. The pixel-level detection results of post-disaster building damage in the Haidi and Changning study areas, as implemented by the proposed method, are illustrated in Figs. 9 and 10. These results showcase the detection outcomes for all building damage within three test zones in each of the affected areas. The first and second columns display the pre- and post-disaster HRRS images, respectively, while the third column shows the ground truth for building damage, and the fourth column presents the pixel-level building damage detection results obtained through our method. In the visual results, the red areas represent correctly identified building damage (TPs), the blue areas denote other objects incorrectly identified as building damage (FPs), the yellow areas indicate building damage misclassified as other objects (FNs), and the white areas signify the background (TNs). From Figs. 9 and 10, it can be observed that the building damage detection outcomes for different test zones closely align with the ground truth, showing no significant discrepancies. This indicates that the building damage detection results achieved with the proposed method are highly reliable and usable.

In addition to the qualitative analysis, we also conduct a quantitative evaluation of the pixel-level damage detection results for each test image in the Haidi and Changning study areas. The quantitative outcomes are recorded in Tables II and III. In the Haidi study area, the overall building

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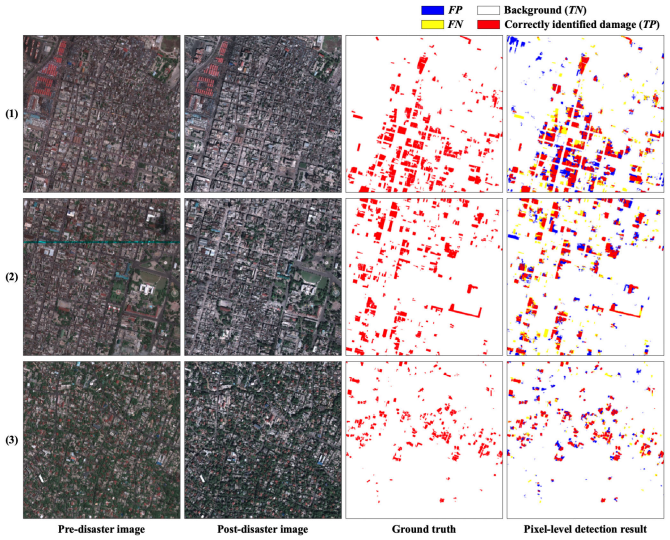


Fig. 9. Building damage detection results in three test images of Haidi study area by our proposed method.

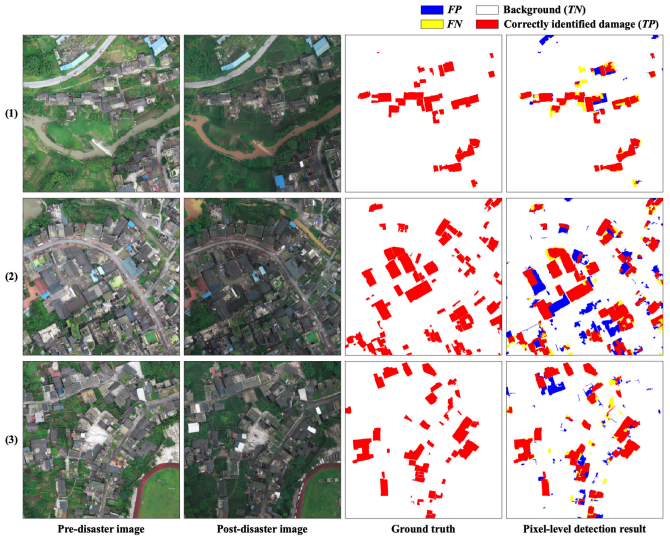


Fig. 10. Building damage detection results in three test images of Changning study area by our proposed method.

damage detection results for all test zones achieve precision, recall, F_1, MIoU, and OA scores of 66.88%, 70.19%, 68.50%, 73.19%, and 94.63%, respectively. However, the quantitative results for test zones 1 and 2 are relatively lower, likely due to the high density of buildings in these areas. In the Changning study area, the overall building damage detection results for all test zones reach precision, recall, F_1, MIoU, and OA scores of 71.97%, 83.61%, 77.35%, 78.48%, and 94.46%, respectively, with the quantitative outcomes showing a notable improvement over the Haidi study area. This improvement can be attributed to the higher quality of UAV imagery in the Changning study area compared with the satellite imagery in the Haidi study area, which offers a higher spatial resolution. In addition, the building density in the Changning study area is significantly lower than in the Haidi study area, which further contributes to the enhanced accuracy of building damage detection.

In the overall methodological framework of the proposed method, pixel-level building damage detection results are derived from scene-level building damage assessment.

TABLE II

PRECISION (%), RECALL (%), F_1 (%), MIOU (%), AND OA (%) OF OUR PROPOSED METHOD FOR TEST IMAGES OF THREE SCENARIOS IN HAIDI

Test Region	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
1	66.02	70.79	68.32	72.01	92.75
2	66.94	67.05	66.99	71.84	93.74
3	68.85	75.45	72.00	76.78	97.40
Three test regions as a whole	66.88	70.19	68.50	73.19	94.63

TABLE III

PRECISION (%), RECALL (%), F_1 (%), MIOU (%), AND OA (%) OF OUR PROPOSED METHOD FOR TEST IMAGES OF THREE SCENARIOS IN CHANGNING

Test Region	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
1	83.57	76.58	79.92	81.81	97.22
2	66.36	85.56	74.75	74.36	90.57
3	75.46	85.42	80.13	81.01	95.60
Three test regions as a whole	71.97	83.61	77.35	78.48	94.46

TABLE IV

QUANTITATIVE RESULTS OF SCENE-LEVEL BUILDING DAMAGE DETECTION ON HAIDI AND CHANGNING STUDY AREAS

Study areas	Evaluation metrics				
	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
Haidi	95.20	86.00	90.37	83.17	90.83
Changning	92.71	84.76	88.56	85.11	93.20

Therefore, the performance of scene-level building damage detection directly influences the pixel-level detection results, playing a crucial role within the entire methodological framework. Given this, it is necessary to analyze the proposed bitemporal scene change detection model (SLgViT) used for scene-level building damage detection, including its training process and output results. Fig. 11 illustrates the convergence process of accuracy and loss, the learning rate update process and the confusion matrices for the proposed SLgViT applied to the Haidi and Changning datasets. For the training of SLgViT on these two datasets, the overall convergence process of accuracy and loss is normal, showing a fast and stable trend. Although there are some fluctuations in valid accuracy and valid loss for the Haidi data during the initial stages of training, the model reaches a basically stable state after more than 20 epochs. This indicates that our SLgViT can achieve stable model training across different study areas.

Table IV presents the quantitative results of scene-level building damage detection in the Haidi and Changning study areas, documenting the evaluation metric scores for the building damage scene category. It is evident from the table that the detection accuracy for the building damage scene category reaches a relatively high level, with each evaluation metric score exceeding 80% in both study areas, thus demonstrating the effectiveness of the proposed SLgViT for scene-level building damage detection. However, upon examining the confusion matrix results from Fig. 11, it is also apparent that the misclassification rate for the building damage scene category is considerably higher than that for the intact building scene category, highlighting the increased difficulty of building

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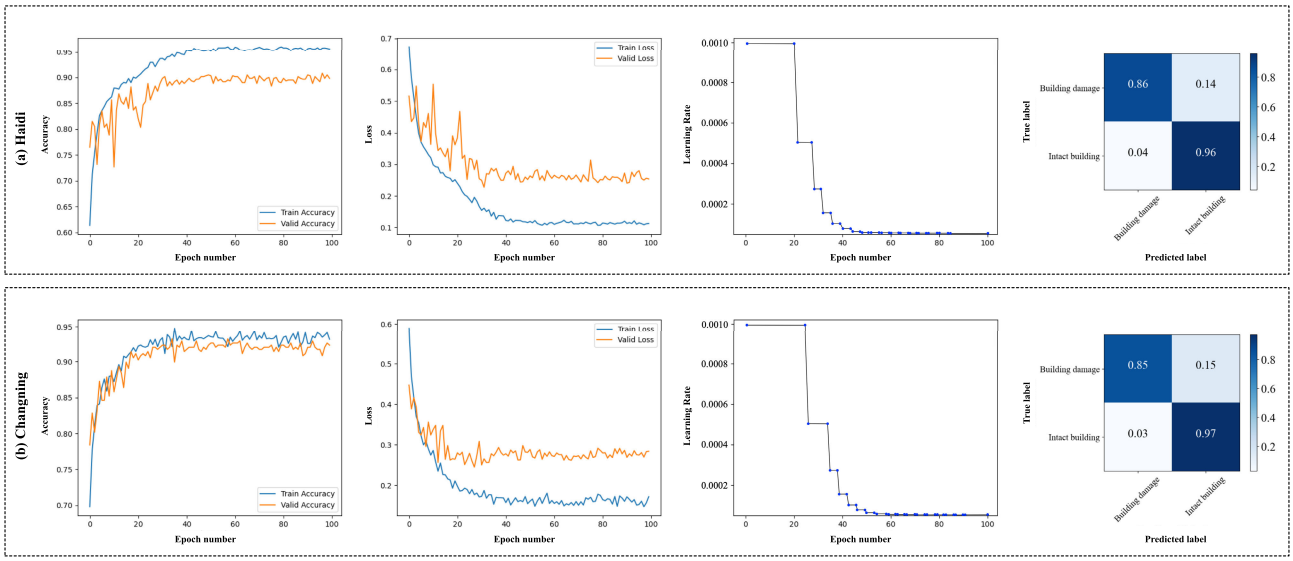


Fig. 11. Convergence process of accuracy and loss, learning rate update process, and confusion matrices of our method on Haidi and Changning datasets. (a) Haidi dataset. (b) Changning dataset.

TABLE V
CATEGORIZATION AND SUMMARY OF THE COMPARISON METHODS

Pixel-level Building Damage Detection			Scene-level Building Damage Detection
Methods	Strategy	Image utilization	Methods
sw_SqueezeNet [45]	Sliding window strategy	Post-disaster images	ViT [8]
sw_ResNet-101 [52]			PiT [9]
sw_T2TViT [12]			CrossViT [10]
sw_MaxViT [19]			TwinsSVT [11]
SEAM [49]	CAM strategy		T2TViT [12]
MDC [48]			NesT [13]
WSAN [50]			ScalableViT [14]
MSD-SCR-Net [6]			CvT [17]
CSSCNN [51]	Pre-disaster image Superpixel segmentation	Pre- and post-disaster images	RegionViT [15]
TSCNN [28]	Pre-disaster building extraction		CCT [18]
FMCNN [27]	Pre-disaster building footprint map		CrossFormer [16]
U-Net [61]	Fully supervised semantic segmentation		MaxViT [19]

damage detection compared with the detection of general objects.

C. Comparison of Pixel-Level Building Damage Detection With Existing Deep Learning Methods

To validate the effectiveness and advantages of the proposed method for pixel-level building damage detection, we conduct both quantitative and qualitative comparisons with recent deep learning methods, which are categorized and summarized in Table V. We compare methods that solely utilize post-disaster images and those that utilize both pre- and post-disaster images. For the former, we select two classic CNN classification networks and two state-of-the-art ViT networks, namely, SqueezeNet [45], ResNet-101 [52], T2TViT [12], and MaxViT [19], where MaxViT integrates CNN and Transformer into a hybrid structure. For these four methods, we employ a sliding window strategy [75] to obtain pixel-level building damage detection results, referred to as sw_SqueezeNet, sw_ResNet-101, sw_T2TViT, and sw_MaxViT. Furthermore, four state-of-the-art WSSS methods (i.e., SEAM [49], MDC [48], WSAN [50], and MSD-SCR-Net [6]) based on CAM are also chosen as comparative methods, similarly utilizing only post-disaster images.

For methods utilizing both pre- and post-disaster images, we select four representative methods employing different strategies.

1) Combining CNN models with pre-disaster HRRS image superpixel segmentation [51], where we replace the DeepLab v2 semantic segmentation network with a ResNet-101 scene classification network, termed as CSSCNN.
2) First extracting pre-disaster buildings, then detecting building damage based on the location of these pre-disaster buildings [28], referred to as the two-stage CNN

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model (TSCNN), where we use FCN instead of CNN to enhance the accuracy of extracting building areas from pre-disaster HRRS images.

3) Utilizing pre-disaster building footprint maps to assist in identifying building damage from post-disaster HRRS images based on an improved SqueezeNet model [27], referred to as FMCNN.
4) A fully supervised semantic segmentation method that cascades pre- and post-disaster HRRS images, referred to as U-Net [61], is chosen to evaluate the accuracy gap between our proposed weakly supervised method and the fully supervised method.

This evaluation serves to validate the comparability of our proposed weakly supervised approach with fully supervised methods, reflecting the effectiveness and advantages of our method from an alternative perspective. It is worth noting that due to the sparse distribution of buildings in the Changning area, the post-disaster building damage data are insufficient to support the training requirements of the fully supervised semantic segmentation method cascading pre- and post-disaster HRRS images. Therefore, we only apply this method in the Haidi area. Within the Haidi study area, we conduct training and validation for the U-Net model using nine concatenated pre- and post-disaster HRRS images, each measuring 3500 × 3500 pixels, along with their corresponding pixel-level building damage labels. Apart from the U-Net model, all the aforementioned comparative methods rely on image-level annotations of building damage for model training.

The quantitative results of pixel-level building damage detection using different methods are listed in Tables VI and VII. From the quantitative results, it is evident that, except for the fully supervised semantic segmentation method based on the U-Net model, our proposed method outperforms other weakly supervised comparative methods in terms of F_1, MIoU, and OA metrics. In the Haidi study area, regarding the recall metric, our proposed method slightly lags behind the SEAM method. However, it can be observed that the SEAM method exhibits poor performance in other metrics, indicating that the SEAM method might mistakenly detect numerous building damage targets and other targets, thus achieving a high recall score. Compared with the FMCNN method, our proposed method shows a decrease in precision by 1.93%. However, there are increases in recall, F_1, MIoU, and OA by 14.25%, 6.79%, 3.9%, and 0.4%, respectively. In the Changning study area, FMCNN exhibits a significant advantage over our proposed method in terms of recall scores. However, it can be noted that there are gaps of 15.66%, 7.16%, 6.44%, and 3.41% in precision, F_1, MIoU, and OA, respectively, when compared with our method. In addition, FMCNN utilizes building footprint maps for assistance, which greatly aids in improving the accuracy of building damage detection. In contrast, our method does not rely on such supplementary data. This confirms the superiority of our proposed method compared with existing weakly supervised comparative methods.

Furthermore, although the fully supervised semantic segmentation method using the U-Net model surpasses our proposed method in all evaluation metrics, it is observed that the differences in each metric are not substantial. This also reflects that despite using weakly supervised training data, the results obtained by our proposed method are commendable. While fully supervised methods have certain advantages in quantitative results, the data preparation required is quite challenging and time consuming. Moreover, for areas with limited data availability, such as the Changning area, obtaining sufficient training data becomes impractical for such methods.

The visual representation of pixel-level building damage detection is crucial because it directly reflects the extent of the disaster and the distribution of building damage, providing essential guidance for rescue and post-disaster recovery efforts. Aside from the semantic segmentation method utilizing the U-Net model, Figs. 12 and 13 illustrate examples of pixel-wise detection results for building damage using various methods in the Haidi and Changning study areas, respectively. From the visual results of different methods, it is apparent that those incorporating pre-disaster imagery or data yield more reliable and accurate pixel-wise building damage detection results. For methods implementing pixel-level building damage detection using the sliding window strategy, such as sw_SqueezeNet and sw_ResNet-101, the visual results tend to be coarse, exhibiting jagged edges due to the sliding window

TABLE VI
QUANTITATIVE COMPARISON OF VARIOUS METHODS ON HAIDI DATASET

Methods	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
sw_SqueezeNet	29.66	71.75	41.97	54.51	83.50
sw_ResNet-101	33.41	62.12	43.45	56.79	86.55
sw_T2TViT	35.17	62.91	45.12	57.85	87.28
sw_MaxViT	37.58	61.67	46.70	59.07	88.30
SEAM	32.62	**72.00**	44.90	56.66	85.31
MDC	40.71	60.32	48.61	60.47	89.40
WSAN	38.25	69.24	49.27	60.06	88.15
MSD-SCR-Net	44.99	55.01	49.49	61.55	90.67
CSSCNN	33.90	63.23	44.14	57.14	86.69
TSCNN	57.54	58.99	58.25	66.85	92.97
FMCNN	**68.81**	55.94	61.71	69.29	94.23
SLgViT	66.88	70.19	**68.50**	**73.19**	**94.63**
U-Net	78.52	74.60	76.51	78.95	96.19

TABLE VII
QUANTITATIVE COMPARISON OF VARIOUS METHODS ON CHANGNING DATASET

Methods	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
sw_SqueezeNet	28.43	77.15	41.54	49.66	75.45
sw_ResNet-101	36.82	76.12	49.63	56.95	82.53
sw_T2TViT	35.81	77.57	49.00	56.22	81.74
sw_MaxViT	44.12	71.05	54.44	61.39	86.55
SEAM	54.87	63.19	58.74	65.38	89.96
MDC	57.78	55.36	56.55	64.58	90.38
WSAN	58.00	54.96	56.44	64.54	90.41
MSD-SCR-Net	53.65	70.84	61.06	66.42	89.78
CSSCNN	38.38	77.80	51.40	58.17	83.36
TSCNN	59.18	84.21	69.51	72.02	91.65
FMCNN	56.31	**93.16**	70.19	72.04	91.05
SLgViT	**71.97**	83.61	**77.35**	**78.48**	**94.46**

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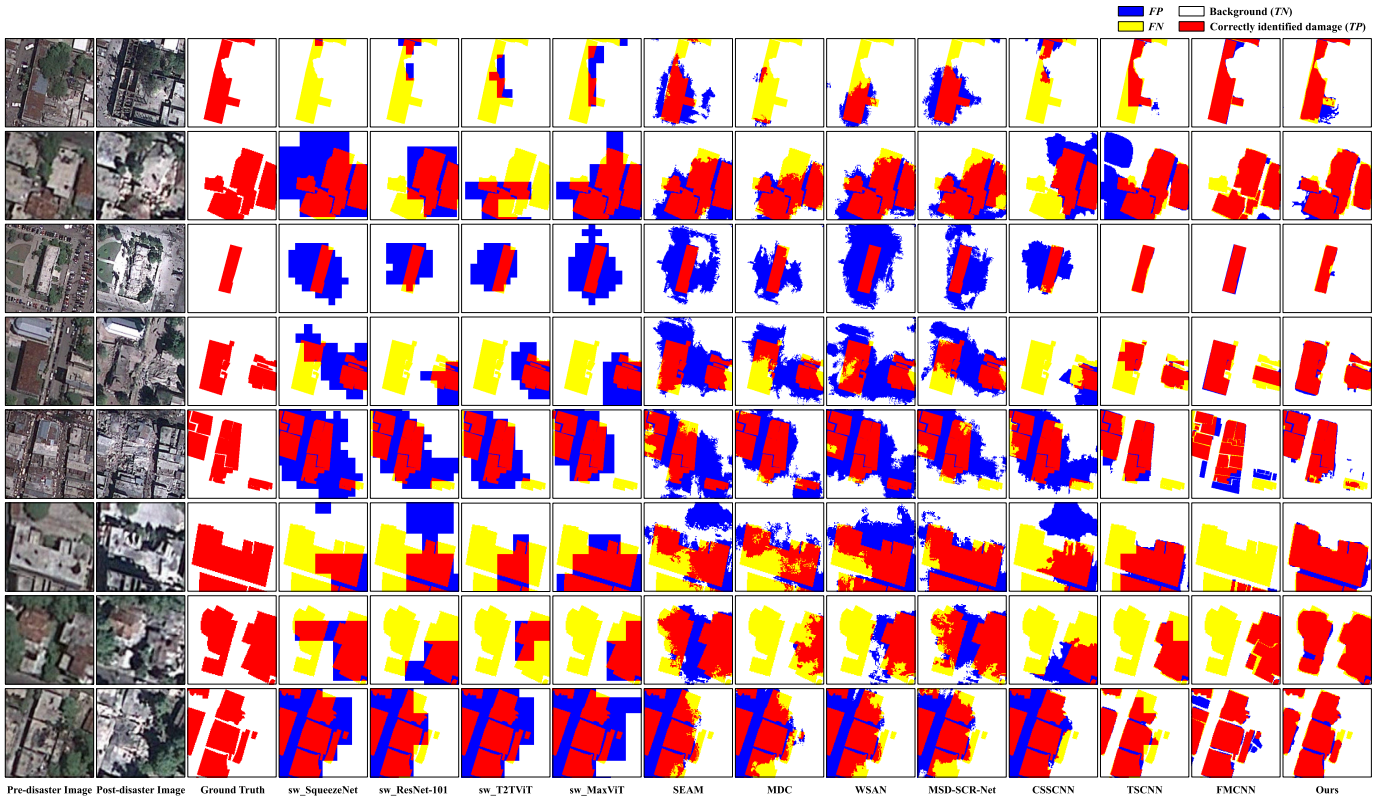


Fig. 12. Examples of pixel-wise detection results of building damage using various methods for the Haidi study area.

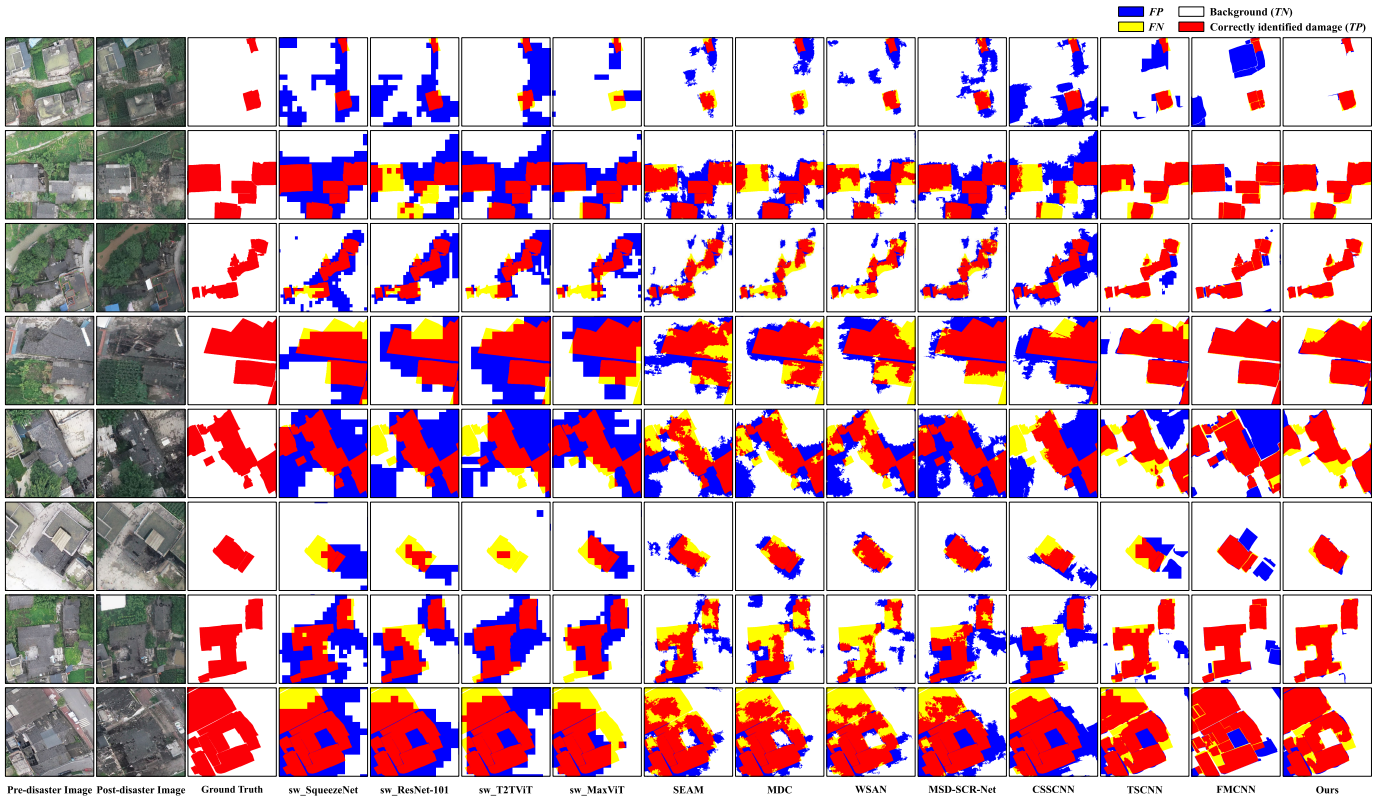


Fig. 13. Examples of pixel-wise detection results of building damage using various methods for the Changning study area.

effect. Despite employing state-of-the-art transformer network architectures, methods such as sw_T2TViT and sw_MaxViT, still fail to achieve substantial improvements in pixel-level detection results, highlighting the limitations of the sliding

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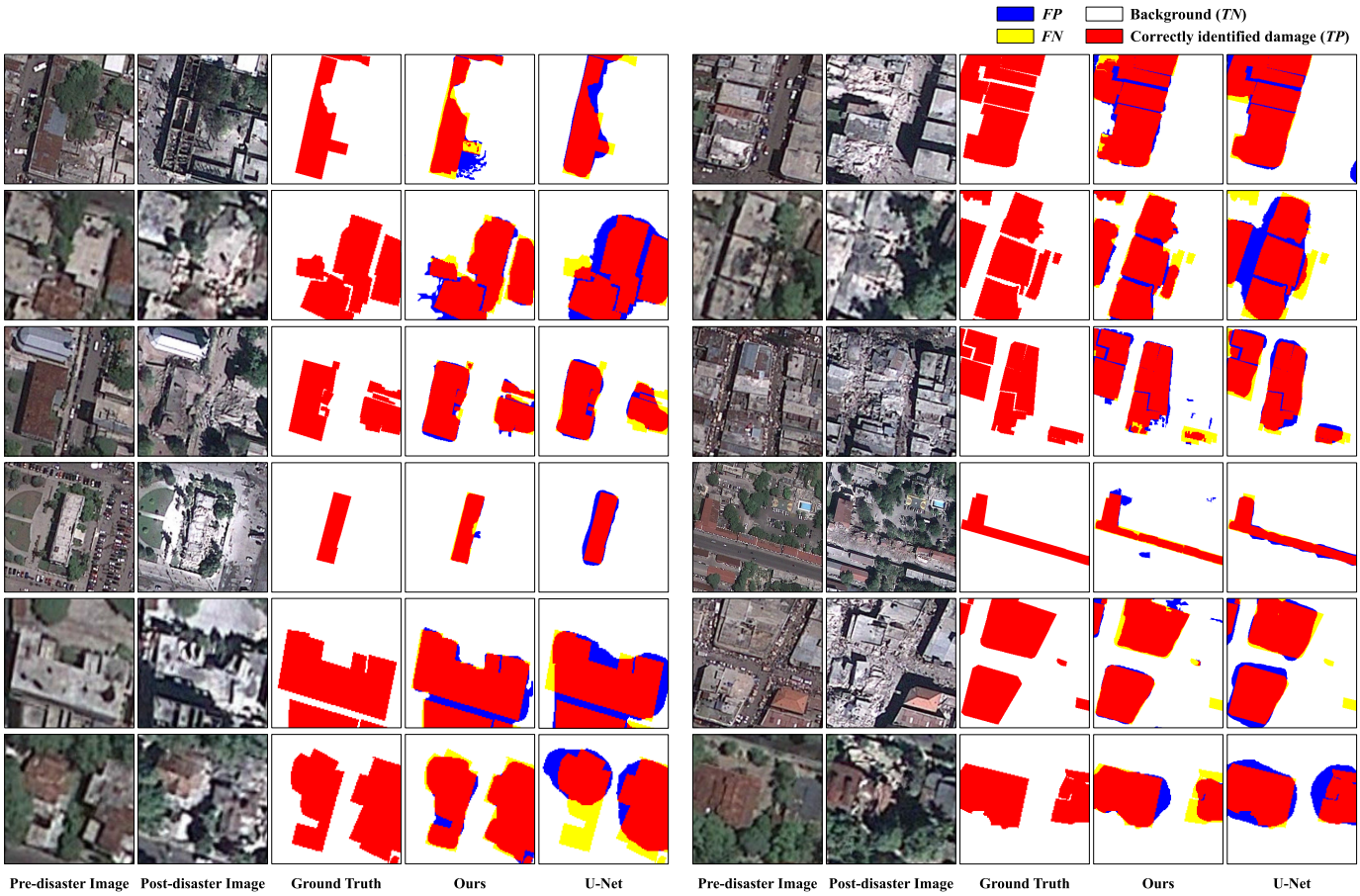


Fig. 14. Comparison of pixel-wise building damage detection results using the U-Net semantic segmentation method for the Haidi study area.

window strategy. Pixel-level detection results obtained from WSSS methods utilizing CAM, such as SEAM, MDC, WSAN, and MSD-SCR-Net, exhibit clearer boundaries in detecting building damage, reflecting finer-grained detection outcomes. Moreover, compared with SEAM, MDC, and WSAN, it can be observed that MSD-SCR-Net, which is specifically designed with the special characteristics of damaged buildings in mind, achieves more comprehensive and consistent results. The extraction of building damage is more homogeneous with reduced internal noise and exhibits clearer boundaries, as illustrated in rows 7 and 8 of Fig. 12 and rows 2 and 3 of Fig. 13. However, when compared with the ground truth, these detection results, while generally depicting contiguous areas of building damage, lack finer details and often fail to accurately pinpoint the specific buildings affected by the damage. The CSSCNN method manages to alleviate some of the issues related to rough results and excessive noise to some extent, demonstrating the intuitive benefits of utilizing pre-disaster imagery and superpixel segmentation. Among all the comparative methods, the TSCNN and FMCNN approaches demonstrate noticeable improvements, providing clear boundaries in the detected building damage and effectively capturing details of damage to individual buildings.

Compared with these comparative methods, the proposed method achieves the best performance in pixel-level detection of building damage and significantly outperforms other methods. Our proposed method generates more accurate and semantically consistent building damage detection results, as evidenced by the results in the 7th row of Fig. 12 and the eighth row of Fig. 13. This improvement is attributed to the design of the network and the systematic methodological framework. Furthermore, when comparing the building damage detection results between the Haidi and Changning study areas, it is observed that the detection results for the Changning area are more accurate and reflect more detailed damage situations, primarily due to the higher quality of HRRS imagery in the Changning area.

Fig. 14 presents a comparison between the pixel-level building damage detection results of our proposed weakly supervised method and the fully supervised U-Net semantic segmentation method. From this visual comparison, it is evident that our weakly supervised approach, despite its less pixel-level rigorous supervision, does not exhibit a significant visual discrepancy from the fully supervised method. Indeed, the results of the fully supervised U-Net method demonstrate a certain advantage in terms of overall object semantic integrity, resulting in smoother visual outcomes, as shown in rows 1 and 4 of Fig. 14. However, it is also observed that, in some detailed aspects, the fully supervised U-Net method may not distinctly delineate the damaged areas of buildings. It sometimes erroneously includes roads and other nonbuilding features in the detection results, as shown in row 2 of Fig. 14.

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TABLE VIII
SCENE-LEVEL BUILDING DAMAGE DETECTION RESULTS OF VARIOUS METHODS ON HAIDI DATASET AND CHANGNING DATASET

Methods	Haidi					Changning				
	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
ViT	64.05	65.33	64.69	47.42	64.33	68.67	54.29	60.64	58.59	78.11
PiT	71.58	69.67	70.61	55.03	71.00	59.46	20.95	30.99	43.66	71.01
CrossViT	72.18	68.33	70.21	55.02	71.00	66.97	69.52	68.22	63.06	79.88
TwinsSVT	79.93	77.00	78.44	65.05	78.83	74.42	60.95	67.02	63.70	81.36
T2TViT	76.90	87.67	81.93	67.49	80.67	79.75	60.00	68.48	65.49	82.84
NesT	85.50	74.67	79.72	67.98	81.00	75.86	62.86	68.75	65.16	82.25
ScalableViT	83.11	83.67	83.39	71.43	83.33	54.29	36.19	43.43	47.37	70.71
CvT	84.51	83.67	84.09	72.66	84.17	73.68	53.33	61.88	60.17	79.59
RegionViT	86.81	83.33	85.03	74.41	85.33	62.50	42.86	50.85	52.20	74.26
CCT	89.05	84.00	86.45	76.72	86.83	68.57	45.71	54.86	55.28	76.63
CrossFormer	86.41	**89.00**	87.68	77.77	87.50	62.12	39.05	47.95	50.79	73.67
MaxViT	91.37	84.67	87.89	79.08	88.33	90.36	71.43	79.79	75.96	88.76
LgViTNet	90.07	84.67	87.29	78.03	87.67	80.56	82.86	81.69	76.75	88.46
SLgViT	**95.20**	86.00	**90.37**	**83.17**	**90.83**	**92.71**	**84.76**	**88.56**	**85.11**	**93.20**

In contrast, the pixel-wise building damage detection results obtained from our method can achieve clearer boundaries of building damage in some cases, aligning more closely with the ground truth. This leads to more advantageous detection outcomes for practical applications, as illustrated in rows 3 and 6 of Fig. 14. Notably, the dense urban structure of the Haidi area, with its abundant building damage samples and pixel-level annotations, satisfies the training requirements of the fully supervised method. However, such conditions are challenging to replicate in real-world disaster mitigation tasks. Therefore, our weakly supervised approach represents a highly valuable alternative.

D. Comparison of Scene-Level Building Damage Detection With State-of-the-Art Methods

The most essential and critical component of our proposed weakly supervised methodological framework for pixel-level building damage detection is the implementation of scene-level detection for building damage through bitemporal scene change detection using the SLgViT network. The effectiveness of scene-level building damage detection is directly related to the accuracy of pixel-level damage detection in buildings. Within our methodological framework, this component can be substituted with any advanced scene classification technique. Therefore, it is crucial to evaluate the effectiveness and advantages of the proposed SLgViT network in comparison with state-of-the-art methods.

To validate the superiority of the proposed SLgViT network, we compare it against state-of-the-art scene classification methods using image-level annotation data from the Haiti and Changning datasets. The methods compared include ViT [8], PiT [9], CrossViT [10], TwinsSVT [11], T2TViT [12], NesT [13], ScalableViT [14], CvT [17], RegionViT [15], CCT [18], CrossFormer [16], and MaxViT [19], as shown in Table V. Among these, CvT, CCT, and MaxViT utilize a hybrid CNN and transformer structure, while the rest are based purely on ViT architecture. In addition, we also test the performance of LgViTNet in scene-level building damage detection to thoroughly assess the effectiveness of the proposed network structure. The scene-level building damage detection results for various methods on the Haidi and Changning datasets are summarized in Table VIII.

Results from Table VIII indicate that our proposed SLgViT network significantly outperforms other methods across all evaluation metrics for scene-level building damage detection in both the Haiti and Changning datasets. Remarkably, the LgViTNet alone also achieves a high level of detection accuracy, particularly in the Changning dataset, where its scene-level detection performance is only surpassed by the SLgViT network. This clearly demonstrates that our scene-level building damage detection network offers superior advantages compared with state-of-the-art methods, making it highly suitable for the proposed weakly supervised methodological framework for pixel-level building damage detection.

V. DISCUSSION

A. Ablation Study

To verify the effectiveness of the proposed LgViT and CSIF modules, we conduct ablation experiments on the Haidi and Changning datasets. For ease of comparison and to more intuitively demonstrate the impact of each module, we remove the LgViT module and the CSIF module from the final SLgViT design to serve as the baseline, noting that removing the CSIF module entails dropping the feature extraction pathway for pre-disaster imagery. Table IX reports the evaluation results of scene-level building damage detection for methods with different module combinations, including baseline, baseline + LgViT, baseline + CSIF, and baseline + LgViT + CSIF. It is evident that adding either the LgViT or CSIF module individually can have a positive impact on the baseline's performance. Notably, the combination of baseline + LgViT + CSIF, i.e., the proposed SLgViT, achieves the best results in all

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TABLE IX

SCENE-LEVEL BUILDING DAMAGE DETECTION RESULTS FROM ABLATION STUDIES ON THE LGVIT MODULE AND THE CSIF MODULE USING HAIDI DATASET AND CHANGNING DATASET

Methods	Haidi					Changning				
	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
Baseline	89.49	76.67	82.59	72.06	83.83	71.07	81.90	76.11	70.00	84.02
Baseline + LgViT	90.07	84.67	87.29	78.03	87.67	80.56	82.86	81.69	76.75	88.46
Baseline + CSIF	89.68	84.00	86.75	77.23	87.17	86.73	80.95	83.74	79.50	90.24
Baseline + LgViT + CSIF	**95.20**	**86.00**	**90.37**	**83.17**	**90.83**	**92.71**	**84.76**	**88.56**	**85.11**	**93.20**

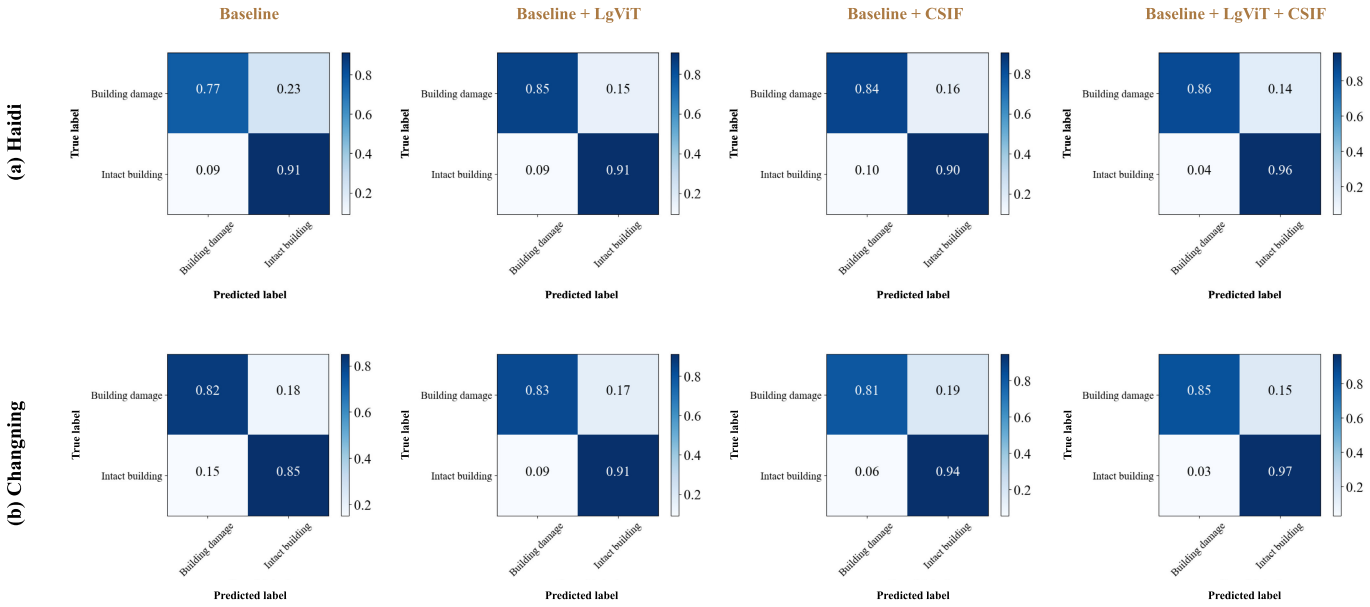


Fig. 15. Confusion matrix comparison of ablation studies on Haidi dataset and Changning dataset.

TABLE X

SCENE-LEVEL BUILDING DAMAGE DETECTION RESULTS FROM ABLATION STUDIES ON LOCAL AND GLOBAL DEPENDENCE MODELING WITHIN THE LGVIT MODULE USING HAIDI DATASET AND CHANGNING DATASET

Methods	Haidi					Changning				
	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
LgViTNet (non-global)	87.64	80.33	83.83	73.13	84.50	79.00	75.23	77.07	72.28	86.09
LgViTNet (non-local)	87.97	85.33	86.63	76.73	86.83	83.51	77.14	80.20	75.69	88.17
LgViTNet	90.07	84.67	87.29	78.03	87.67	80.56	82.86	81.69	76.75	88.46
SLgViT (non-global)	91.04	84.67	87.74	78.82	88.17	87.00	82.86	84.88	80.69	90.83
SLgViT (non-local)	91.99	**88.00**	89.95	82.09	90.17	92.55	82.86	87.44	83.86	92.60
SLgViT	**95.20**	86.00	**90.37**	**83.17**	**90.83**	**92.71**	**84.76**	**88.56**	**85.11**	**93.20**

metrics for building damage scene detection on both datasets, reflecting the high effectiveness and reasonable combination of the proposed LgViT and CSIF modules.

To explain the sources of error in the methods with different module combinations, we illustrate their confusion matrices in Fig. 15. It can be observed that for the Haidi dataset, the baseline method primarily misclassified many building damage scene categories as intact building scene categories, while both the LgViT and CSIF modules can effectively mitigate this issue. In the tests on the Changning dataset, the performance gap of the baseline method is mainly manifested in the confusion between building damage scene categories and intact building scene categories, while the LgViT and CSIF modules significantly improve the detection accuracy for intact building scene categories. In the tests on both datasets, our proposed method significantly improves detection accuracy for both building damage scene categories and intact building scene categories by integrating the LgViT and CSIF modules, further highlighting the effectiveness of the proposed LgViT and CSIF modules.

In addition, we conduct ablation experiments to assess the impact of local and global information extraction on network effectiveness. Table X presents the scene-level building damage detection results from ablation studies on local

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and global dependence modeling within the LgViT module using the Haidi and Changning datasets. In these experiments, the proposed LgViTNet and SLgViT represent the full models incorporating both local and global dependence modeling processes. To analyze the effects of each component, we create variations by removing either the global or local dependence modeling. Specifically, we remove the global dependence modeling component from LgViTNet and SLgViT, while retaining the local dependence modeling, resulting in LgViTNet (non-global) and SLgViT (non-global). Conversely, we remove the local dependence modeling component while retaining the global dependence modeling, resulting in LgViTNet (non-local) and SLgViT (non-local). These ablation settings are used to evaluate the impact of local and global information extraction.

The evaluation results presented in Table X demonstrate that LgViTNet, with both local and global dependence modeling, significantly outperforms the networks lacking either local or global dependence modeling [i.e., LgViTNet (non-global) and LgViTNet (non-local)]. Similarly, SLgViT also shows significantly better performance compared with the network settings of SLgViT (non-global) and SLgViT (non-local). These results indicate that considering both local and global information extraction yields greater performance benefits for building damage detection models compared to focusing solely on either local or global information extraction. This also fully demonstrates the rationality and effectiveness of the combined design of local dependence modeling and global dependence modeling processes in the proposed LgViT module for extracting local and global information. Moreover, compared with baseline models of LgViTNet and SLgViT without local and global dependence modeling (i.e., the baseline and baseline + CSIF, as shown in Table IX), networks incorporating either local or global dependence modeling [i.e., LgViTNet (non-global), LgViTNet (non-local), SLgViT (non-global), and SLgViT (non-local)] show improved effectiveness. This verifies that both local and global dependence modeling, which considers the extraction of local and global information, positively impacts performance. Notably, the inclusion of only local dependence modeling results in a smaller improvement compared with baseline models. This might be due to the baseline models already paying some attention to local information. However, the process of local dependence modeling is indispensable for the proposed LgViT module. Its combination with global dependence modeling is crucial for achieving the best results.

Furthermore, to evaluate the impact of different ablation settings of the network on the final pixel-wise detection results of building damage, we conduct a visual analysis based on the proposed building damage detection framework and various ablation settings. Fig. 16 shows the examples of pixel-wise detection results using various ablation settings for the Haidi and Changning study areas. In generating the detection results, the remaining settings of the proposed methodological framework are kept consistent, altering only the scene-level building damage detection network settings. This allows for a more intuitive analysis of how different ablation settings affect the final pixel-wise detection results of building damage. The visual examples of pixel-wise detection results in Fig. 16 reveal that the baseline detection results contain numerous false identifications of nonbuilding damage and fail to detect many instances of actual building damage. Incorporating the LgViT or CSIF modules into the baseline configurations, i.e., baseline + LgViT and baseline + CSIF, significantly mitigates the issues observed in the baseline detection results. This demonstrates that both modules positively impact performance. However, the baseline + LgViT configuration lacks the assistance of pre-disaster information, and the baseline + CSIF configuration does not consider the local and global information provided by the LgViT module. As a result, many false detections remain unavoidable, and their performance falls significantly short compared with the proposed approach (baseline + LgViT + CSIF). The LgViTNet (non-global) and SLgViT (non-global) configurations show some improvement in the visual results of building damage detection by incorporating local dependence modeling. However, these improvements are not significant, possibly because the base network is already effective at extracting and utilizing local information. In contrast, the LgViTNet (non-local) and SLgViT (non-local) configurations, which incorporate global dependence modeling, significantly improve the performance of building damage detection, particularly in providing a more complete and comprehensive identification of building damage objects. This highlights the effectiveness of global dependence modeling within the LgViT module and reflects the inherent weakness of the base network in processing global information. We notice that the proposed approach performs best, delivering the most complete and accurate pixel-wise building damage detection results compared with the other network ablation settings. This demonstrates that the designed LgViT and CSIF modules, as well as the local and global dependence modeling within the LgViT module, are indispensable components for enhancing the network's performance in building damage detection.

B. Performance Analysis of Siamese Model Fusion Architecture

For the performance analysis of the Siamese model fusion architecture, we test twin-tower concatenate (TTC), twin-tower subtract (TTS), and our proposed CSIF on the Haidi and Changning datasets. TTC and TTS are the two classic and widely used Siamese model fusion architectures. TTC concatenates features extracted from bitemporal image scenes along the channel dimension, while TTS differs from TTC by subtracting the extracted features elementwise instead of concatenating them. We select T2TViT, MaxViT, and our proposed LgViTNet as foundational models for experimentation, as T2TViT and MaxViT represent pure ViTs and hybrids of ViTs with CNNs, respectively.

The experimental results are summarized in Table XI. Overall, adopting the Siamese bitemporal building damage detection architecture positively affects the models, demonstrating that comparing buildings and their surrounding environments in post-disaster images with those in pre-disaster images can extract more valuable information. It has been observed that the TTS strategy generally outperforms the TTC

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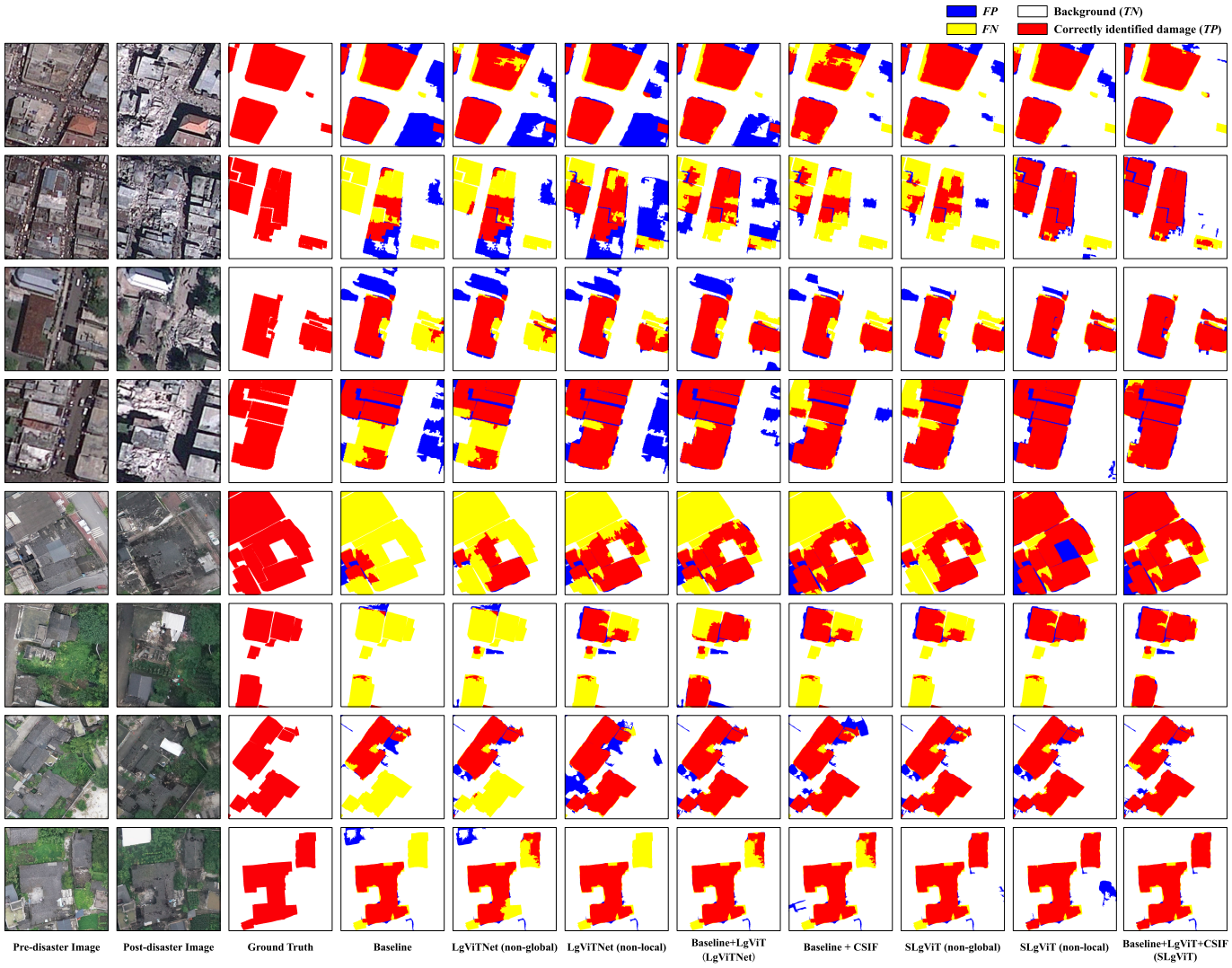


Fig. 16. Examples of pixel-wise detection results using various ablation settings for the Haidi and Changning study areas. The first-to-fourth rows correspond to the Haidi study area, and the fifth-to-eighth rows correspond to the Changning study area.

TABLE XI

PERFORMANCE COMPARISON OF DIFFERENT SIAMESE MODEL FUSION ARCHITECTURES ON HAIDI DATASET AND CHANGNING DATASET

Methods	Haidi					Changning				
	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)	Precision (%)	Recall (%)	F_1 (%)	MIoU (%)	OA (%)
T2TViT	76.90	87.67	81.93	67.49	80.67	79.75	60.00	68.48	65.49	82.84
T2TViT + TTC	90.87	79.67	84.90	75.11	85.83	72.64	73.33	72.99	67.81	83.14
T2TViT + TTS	85.76	90.33	87.99	78.03	87.67	76.00	72.38	74.15	69.34	84.32
T2TViT + CSIF	89.86	88.67	89.26	80.72	89.33	71.54	**88.57**	79.15	72.75	85.50
MaxViT	91.37	84.67	87.89	79.08	88.33	90.36	71.43	79.79	75.96	88.76
MaxViT + TTC	90.07	84.67	87.29	78.03	87.67	77.78	66.67	71.79	67.74	83.73
MaxViT + TTS	88.78	87.00	87.88	78.57	88.00	78.10	78.10	78.10	73.05	86.39
MaxViT + CSIF	83.14	**95.33**	88.82	78.48	88.00	89.01	77.14	82.65	78.60	89.94
LgViTNet	90.07	84.67	87.29	78.03	87.67	80.56	82.86	81.69	76.75	88.46
LgViTNet + TTC	77.69	94.00	85.07	71.44	83.50	87.38	85.71	86.54	82.49	91.72
LgViTNet + TTS	93.14	86.00	89.43	81.52	89.83	90.00	77.14	83.08	79.11	90.24
LgViTNet + CSIF	**95.20**	86.00	**90.37**	**83.17**	**90.83**	**92.71**	84.76	**88.56**	**85.11**	**93.20**

strategy in most cases, which may be due to the feature redundancy caused by concatenating features extracted from bitemporal image scenes, making the model less sensitive to feature differences. TTS aims to more directly capture

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TABLE XII
COMPARISON OF SPATIAL AND TEMPORAL COMPLEXITIES FOR DIFFERENT METHODS

Models	Model size (M)	Time Cost (s)
ViT	252.1	8
PiT	187.7	10
CrossViT	218.6	17
TwinsSVT	1310.7	47
T2TViT	92.0	15
NesT	64.6	16
ScalableViT	301.9	50
CvT	70.5	15
RegionViT	49.2	19
CCT	88.1	10
CrossFormer	113.3	43
MaxViT	317.4	38
LgViTNet	8.1	16
SLgViT	23.2	37

the differences between pre-disaster and post-disaster images, a key indicator for building damage assessment, hence achieving better results. Interestingly, LgViTNet + TTC outperforms LgViTNet + TTS on the Changning dataset, suggesting that the TTC strategy has advantages in some scenarios, possibly because the concatenation method can more comprehensively utilize features from pre-disaster and post-disaster images. Comparing the statistical results in Table XI, it is evident that utilizing our proposed CSIF fusion strategy achieves the best performance. This indicates that the CSIF fusion strategy can not only integrate pre- and post-disaster image features comprehensively but also sensitively focus on feature differences, resulting in more accurate building damage detection. Furthermore, it is worth noting that both MaxViT + TTC and MaxViT + TTS show a slight decrease in performance metrics compared with MaxViT alone, reflecting that TTC and TTS may not be very suitable for specific foundational network architectures. However, the results of MaxViT + CSIF still demonstrate higher test scores compared with MaxViT alone, further highlighting the effectiveness and robustness of our proposed CSIF strategy.

C. Spatial and Temporal Complexity Analysis of the Proposed Method

To verify the efficiency of the proposed method, we conduct a comparative analysis of spatial and temporal complexities with other methods. Table XII records the time spent on scene detection for 600 test images of 256 × 256 pixels and the size of each model, based on which the efficiency of different methods is analyzed. In the temporal complexity test, all methods are executed on a single NVIDIA GTX 1080Ti GPU to ensure a fair comparison.

Compared with other methods, our LgViTNet is only 8.1 M in size, and even the Siamese-structured SLgViT model is just 23.2 M, demonstrating our model's significant advantage in spatial complexity. Regarding temporal complexity, our LgViTNet model only requires 16 s to predict scenes from 600 images, which is relatively less time consuming. Due to the pre- and post-disaster bitemporal image scene block detection employed by SLgViT, there is a slight increase in time cost, yet it remains less than that of the TwinsSVT, ScalableViT, CrossFormer, and MaxViT methods. Furthermore, as demonstrated in previous accuracy comparisons, our proposed SLgViT shows a clear advantage. Therefore, sacrificing some temporal complexity is justified. Based on the above analysis, it can be concluded that our proposed SLgViT achieves an optimal balance between efficiency and accuracy, making it an effective and efficient method.

VI. CONCLUSION

Due to the extremely diverse and intricate appearance of damaged buildings in post-disaster HRRS images, coupled with the tendency for their features to be easily confused with surrounding objects, automated assessment of building damage becomes a challenging task. In this article, a significant breakthrough direction in current research lies in fully exploiting the visual information of building boundaries and the deeper semantic information of building scenes contained in pre-disaster images, integrating them with the damage characteristics of buildings in post-disaster images to enhance the capability of automated building damage assessment. In this article, we propose a weakly supervised building damage assessment method based on scene change detection in pre- and post-disaster bitemporal images. This method integrates building object localization, scene change detection, and pixel-level building damage inference into a unified system. Specifically, the method first generates fine-grained sub-building objects with detailed boundaries from pre-disaster images by combining semantic segmentation of buildings with superpixel segmentation. Then, bitemporal image scene blocks obtained using sub-building objects as clues are input into our proposed SLgViT network, enabling scene change detection guided by deep semantic information from pre-disaster images. Finally, the change detection results serve as the basis to depict pixel-level building damage in post-disaster images.

Comprehensive experimental results from the Haiti and Changning datasets indicate that our proposed weakly supervised method, based on pre- and post-disaster bitemporal image scene change detection, exhibits exceptional performance and efficiency in building damage detection tasks. It achieves reliable results in detecting building damage, demonstrating strong applicability. Comparative analysis reveals that leveraging visual information of building boundaries from pre-disaster images significantly mitigates interference from areas unrelated to building damage in post-disaster images, markedly enhancing the accuracy of building damage detection outcomes. Moreover, our findings suggest that a unified model that combines deeper semantic information of building scenes from pre-disaster images with post-disaster building damage characteristics is highly suitable and effective for building damage detection tasks, significantly improving detection performance. In addition, ablation studies confirm the effectiveness of the main components of the proposed SLgViT network, namely, LgViT module and CSIF module.

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We believe that the weakly supervised method presented in this article offers a novel and valuable framework for building damage assessment tasks. It has the potential to serve as a foundational framework for the future research.

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